

GENERAL SPECIFICATION G4.1

Valves, Penstocks and Stoplogs

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G4.1 VALVES, PENSTOCKS AND STOPLOGS

G4.1.1 General

- (a) All valves shall be suitable for use in the specified working environment and application. Generally, the pressure designation shall not be less than PN10 unless otherwise specified.
- (b) The design pressure rating for all valves shall not be less than 1.5 times the connected pump shut-off head, or 1.5 times the pipeline maximum operating pressure or PN10, whichever is greater. Where specific applications exist such that reasons may be elucidated to justify deviation from these requirements, these shall be submitted in writing to the S.O. for review and approval.
- (c) Where necessary, reduction gear drive shall be employed to achieve the required torque for valve operation and restrict the hand wheel diameter to no greater than 450 mm.
- (d) Configuration of the hand wheel shall be that when the hand wheel is rotated in the anti-clockwise direction, it shall open the valve/penstock whilst rotating the hand wheel in clockwise direction shall close the valve/penstock. The direction of rotation shall be clearly cast/embossed and highlighted in prominent colour (red or suitable contrasting colour) on the hand wheel with the wordings 'Close' or 'Open' and directional arrows. The rotation of threaded stems or gearboxes' output shafts shall give the intended opening or closing of valves or penstocks matching the handwheel rotation direction.
- (e) There shall be provisions such as spring washers, lock nuts, cotter pins, etc. to secure handwheels and fasteners from becoming loose due to vibration arising from operation of penstocks/valves.
- (f) Unless otherwise specified, valves shall have integral flanges drilled to BS EN 1092, BS EN 545 or BS EN 598 where applicable. In general, metallic flanges shall be raised face flanges whilst thermoplastic and thermoset plastic flanges as well as mating metallic and non-metallic flanges shall be flat faces. Flanges to other standards shall be used only if approved by the S.O. and provided that any differences do not affect mating dimensions. Back faces of flanges shall be machined.
- (g) Gate valves and butterfly valves shall be suitable for flow in either direction.
- (h) All standard valves shall be suitable for frequent operation and for infrequent operation after long periods in the open or closed condition.
- (i) Rubber used in valves shall be ethylene propylene rubber (EPDM or EPM) or styrene butadiene rubber (SBR). It shall comply with the requirements of Appendix B of BS 6716, be suitable for making a long term flexible seal and be resilient to mechanical stress and resistant to chemical or bacteriological attack leading to deterioration of the flexible seal.
- (j) Where valve bodies or other parts are specified to be of cast iron to BS EN 1561, valves incorporating bodies or other parts of spheroidal graphite iron (ductile iron) to BS EN 1563 may be supplied if available, provided that they otherwise comply with the Specifications and S.O.'s approval.

- (k) Unless otherwise specified or expressly approved by the S.O., valve components shall be of materials not inferior in strength or resistance to corrosion to those listed below:

S/No.	Part Description	Material
(a)	Bodies, bonnets, covers, cowls wedges and plungers	Spheroidal graphite iron to BS EN 1563 grade 500/7 or cast iron to BS EN 1561 grade 220. Plain carbon steel to BS EN 10213 Part 2 Grade 430 BS EN 10293 Steel A1.
(b)	Pistons, sleeves, liners, bushes, seats, rings and seat seals	Bronze to BS EN 1982 grade PB1 guides or gunmetal to BS EN 1982 grade LG2. High tensile brass to BS EN 12420 CP114
(c)	Hinge bushes and pins	Bronze to BS EN 12163 grade PB102.
(d)	Valve stems, spindles, extension spindles and pins	Stainless steel grade 410 S21 to BS EN 10088-3, BS EN 10095, BS EN 10250-4, BS EN 10085, PD 970, BS EN 10087, BS EN 10083-1, BS EN 10084.
(e)	Extension spindles (plain ends)	Stainless steel Grade 316S16 to BS EN 10088-3, BS EN 10095, BS EN 10250-4, BS EN 10085, BS EN 10087, BS EN 10083-1, BS EN 10084.

- (l) Requirements for valves used in chemical applications are given under the corresponding sub-sections of the Specification.
- (m) Where there is contact or exposure to liquids in which chlorides are present, the wetted parts and components of the valves/stoplogs/penstocks shall be manufactured of materials guaranteed against corrosion and wear against chlorides concentration as follows:

Liquid in contact	Chloride concentration limit
Saturated brine	At least 160, 000 mg/l
Seawater	At least 19, 000 mg/l
All other liquid with chlorides present	At least 10, 000 mg/l

- (n) Unless otherwise specified, parts and materials of equipment in contact with seawater shall be fabricated in Super Duplex stainless steel grade SAF2507.

All Super Duplex shall have a minimum Pitting Resistance Equivalent Number (PREN) of 40 and Crevice Factor (CF) above 38 as calculated as follows:

$$PREN = Cr (\%) + 3.3 \times Mo (\%) + 16 \times N (\%)$$

$$CF = Cr (\%) + 3 \times Mo (\%) + 15 \times N (\%)$$

- (o) Valves potentially coming in contact with ozone and ozonated water shall be supplied with ozone resistant gaskets, O-rings and seals. The proposed materials of construction shall be subject to approval.
- (p) All valves of sizes greater than DN300 mm shall be power actuated – electric, pneumatic or hydraulic unless otherwise specified. Process requirements may dictate that valves of sizes 300 mm and smaller require non-manual actuators. These specific requirements are found in the Particular Specifications for the applications of concern. Where valves are installed in chambers below ground level for ease of operation, valves shall be provided with extension spindles including protection covers and headstocks to allow operation from above ground level without causing any obstruction. Headstocks shall be of cast iron and shall be from the standard product range of a reputable manufacturer. The height of the hand wheel above the bottom of the headstock shall not be less than 900 mm and for actuated valves, the actuator shall be installed on the headstock. For valves installed at higher elevations, operation by means of chain shall be provided within the operator's reasonable height. These requirements shall be met irrespective of whether they are shown on the drawings or not.
- (q) Electric actuators for valves of size greater than DN 300 mm shall be rated for not less than 20% more than the torque required to operate the valve.
- (r) Where valves are to be actuated, the valves shall be supplied complete with integral mounting flange for the actuator. Actuators shall be tested, verified with supporting test documentation (inclusive of test records, documents and certificates) and fitted at the valves manufacturer's works. Actuators used for flow control shall be of the modulating type.
- (s) Depending on the application headstocks and their actuators shall be adequately supported either on the civil structure or on substantial fabricated steel supports fixed to the civil structure or other structural steelwork at ground level or above. The Contractor shall be responsible for providing all details of required openings in civil structures to accommodate the extension spindles without causing any obstruction or blockage.
- (t) Where it is necessary to offset the headstock horizontally from the valve spindle, this shall be carried out using two universal joints, one at the bottom of the extension spindle and the other at the top below the headstock. Universal joints shall be from the standard product range of a reputable manufacturer and shall be subject to the approval of the S.O..
- (u) All valves supplied shall have provisions for locking at any random position over the entire operating range.
- (v) Unless otherwise specified, all the supplied valves regardless of whether manually operated type or not shall be provided with indicators to show the percentage (%) of opening closing for the disc positions.
- (w) The Contractor shall ensure the valves provided are suitable for their intended applications.
- (x) Suitable flange adaptors shall be provided in the pipework adjacent to each valve of size DN 300 mm and above, unless otherwise specified.

- (y) Flange adaptors/coupling shall be tied to pipework with appropriate length of rods at each quadrant of pipe and not to pipework supports.

G4.1.2 Reference Standards

- (a) The following is a list of standards which may be referenced in this Section:
- (i) American National Standards Institute (ANSI): C111/A21.11, Rubber-Gasket Joints for Ductile-Iron Pressure Pipe and Fittings.
 - (ii) American Society of Sanitary Engineers (ASSE), Performance Requirements for Hose Connection Vacuum Breakers.
 - (iii) American Society for Testing and Materials (ASTM):

Standard	Subject
A276	Standard Specification for Stainless Steel bars and Shapes
A351	Standard Specification for Castings, Austenitic, Austenitic- Ferric (Duplex), for Pressure-Containing Parts
A564	Standard Specification for Hot-Rolled and Cold-Finished Age-Hardening Stainless-Steel bars and Shapes
B61	Standard Specification for Steam or Valve Bronze Castings
B62	Standard Specification for Composition Bronze or Ounce Metal Castings
B98	Standard Specification for Copper-Silicon Alloy Rod, bar and Shapes
B127	Standard Specification for Nickel-Copper Alloy (UNS N04400) Plate, Sheet and Strip
B139	Standard Specification for Phosphor Bronze Rod, bar and Shapes
B164	Standard Specification for Nickel-Copper Alloy Rod, bar and Wire
B194	Standard Specification for Copper-Beryllium Alloy Plate, Sheet, Strip, and Rolled bar
B584	Standard Specification for Copper Alloy Sand Castings for General Applications
D429	Test Methods for Rubber Property-Adhesion to Rigid Substrates
D1784	Standard Specification for Rigid Poly (Vinyl Chloride) (PVC) Compounds and Chlorinated Poly (Vinyl Chloride).

(iv) American Water Works Association (AWWA)

Standard	Subject
C500	Metal-Seated Gate Valves for Water Supply Service
C504	Rubber-Seated Butterfly Valves
C507	Ball Valves
C508	Swing-Check Valves for Waterworks Service, 2 in. to 24 in. NPS
C509	Resilient-Seated Gate Valves for Water Supply Service
C510	Double Check Valve, Backflow-Preventer Assembly
C511	Reduced-Pressure Principle Backflow Prevention Assembly
C520-14	Knife Gate Valves Sizes, 2 In. (50 mm) Through 96 In. (2, 400 mm)
C540	Power-Actuating Devices for Valves and Sluice Gates
C550	Protective Epoxy Interior Coatings for Valves and Hydrants
C560	Cast-Iron Slide Gates
C606	Grooved and Shouldered Joints
C800	Underground Service Line Valves and Fittings

(v) British Standards Institute (BSI)

Standard	Subject
BS EN 1092	Circular Flanges for Pipes, Valves and Fittings
BS EN 1171	Industrial valves. Cast iron gate valves
BS EN 16767	Industrial valves. Steel and cast iron check valves
BS EN 593	Industrial valves. Metallic butterfly valves
BS EN 13397	Industrial valves. Diaphragm valves made of metallic materials
BS 5158	Specification for Cast Iron Plug Valves
BS 7775	Penstocks for use in water and other liquid flow applications. Specification
BS ISO 7121	Steel ball valves for general-purpose industrial applications

- (vi) International Electro technical Commission (IEC): 60529, Degrees of Protection Provided by Enclosures (IP Code).

- (vii) Manufacturers Standardization Society (MSS):

Standard	Subject
SP-81	Stainless Steel, Bonnet less, Flanged Knife Gate Valves
SP-88	Diaphragm Type Valves

- (viii) Public Utilities Board, Singapore (PUB):

- 1) Public Utilities (Water Supply) Regulations on the Public Utilities Act;
- 2) Public Utilities Board Handbook on Applications for Water Supply; and
- 3) Standards for Water Fittings, Public Utility Board.

- (ix) Singapore Productivity and Standards Board (PSB):

- 1) CP 48, Code of Practice for Water Services;
- 2) Code of Practice for Sanitary Plumbing and Drainage; and
- 3) Product Listing Scheme.

G4.1.3 Submittals

- (a) As minimum requirement, Contractor shall submit the Shop Drawings, Samples and Technical Data including or are associated with the following items:
- (i) Equipment Assembly: Make, model, weight, and kW rating of each component;
 - (ii) Manufacturer's catalogue information, descriptive literature, specifications, dimensional layouts and identification of materials of construction;
 - (iii) Detailed structural, mechanical, and electrical drawings showing equipment fabrications and interface with other items. Include dimensions, size, and locations of connections to other work and weights of associated equipment;
 - (iv) Power and control wiring diagrams, including terminals and numbers;
 - (v) Complete motor nameplate data and tags;
 - (vi) The opening and closing torque and loads for the valve / penstock that will be transmitted to support structure with operator at extreme positions and load;
 - (vii) Sizing calculations for open-close/throttle and modulating valve actuators;
 - (viii) Shaft calculations for each Penstock / valves and service condition;
 - (ix) External utility requirements such as power and drain for each component;
 - (x) Functional description of internal and external instrumentation and controls to be supplied including list of parameters monitored, controlled, or alarmed;
 - (xi) Power and control wiring diagrams, including terminals and numbers;
 - (xii) Field performance test procedures;

- (xiii) Hydraulic power unit manufacturer's list of reference installations showing 5 years of experience with similar systems;
- (xiv) Electric actuator manufacturer's list of reference installations showing 5 years of experience with similar non-proprietary fieldbus network systems as specified;
- (xv) Manufacturer's Certificate of Compliance for materials;
- (xvi) Factory Inspection, technical schedules and Test Reports;
- (xvii) Special shipping, storage and protection, and handling instructions;
- (xviii) Manufacturer's installation instructions;
- (xix) Routine maintenance requirements prior to plant start up;
- (xx) Manufacturer's Certificate of Proper Installation;
- (xxi) Operation and Maintenance Manual;
- (xxii) Certificates of Conformance certifying that materials and equipment supplied conform to the respective standards referenced herein;
- (xxiii) Calibration Certificates certifying that factory test equipment has been calibrated and checked against an approved standard within the last 12 months; and
- (xxiv) Certify that field test equipment has been calibrated and checked against an approved standard within the last 12 months.

G4.1.4 Qualification

- (a) The manufacturer of the valves, penstocks and stoplogs shall have at least 10 years' experience in the design, fabrication and supply of the valves as specified in this specification which include the larger diameter valve up to 3000 mm. Such valves previously supplied for reference projects shall have been in actual operational service for a minimum of 3 years.

G4.1.5 Protection against Climatic Conditions

- (a) The valves, penstocks and stoplogs supplied shall be of the appropriate grade and quality and shall be adequately protected against the climatic conditions at the Site. The Contractor shall take these conditions into account in deciding what grade, quality and protection is required. Any valve, penstock or stoplog found to be unsuitable for use under these conditions shall be removed and replaced by suitable ones by the Contractor at no extra expense to the Board.
- (b) Valve, penstock and stoplog bodies, hatchboxes and all other castings shall be coated for tropical conditions. Where this is not applicable, they shall be thoroughly cleaned and be given two coats of coal tar epoxy with a resin base. A suitable rust inhibitor, such as a high melting point grease of approved quality, shall cover machined surfaces.

- (c) All submerged moving parts, or the pins, spindles and faces, etc. in contact with the fluid, shall be of non-corrodible materials. Any parts that show signs of corrosion or wear during the Defects Liability Period shall be replaced and rectified by non-corrodible material of suitable quality for the purpose at no extra expense to the Board. The replacement materials shall have the samples submitted for approval by S.O. prior to being procured and brought to site. Care shall be exercised in the choice of metals for use in the valves, penstocks or stoplogs to totally eliminate or reduce the effects of bi-metallic corrosion to a minimum. The foregoing shall apply also to the moving parts exposed to the weather.

G4.1.6 Inspection

- (a) All valves/penstocks/stoplogs shall be offered for inspection by the S.O. or Inspectors appointed by the Board at the Contractor's premises or at the place of manufacture. The Contractor shall provide such office facilities, assistance, transport, labour, materials, apparatus and instruments as may be necessary to allow a thorough and extensive inspection to be carried out.
- (b) The S.O. or the Inspectors shall be provided with access to the premises of the Contractor or the manufacturers at all reasonable times, and shall be at liberty to inspect the manufacture of valves/penstocks/stoplogs and related component parts at all stages. The charging of cupolas, melting, drawing off, conveying and pouring of all metals shall be carried out in such a manner and at such time or times that the S.O. or the Inspectors desire to see that the work is carried out in accordance with this Specification, particularly but not limiting to the following requirements and conditions:
 - (i) The manufacture of valves, penstocks, stoplogs and component parts at all stages shall be offered for the S.O.'s inspection by the Contractor. If found defective or inferior in quality to, or differing in form or material from, the requirements of the Contract, such valves/penstocks/stoplogs shall be rejected and replacement shall be made to the approval of the S.O. at no cost to the Board. The whole of any consignment may be rejected if any valve, penstock, stoplog or component part is found not to conform in any respect to the requirements and specifications of the Contract.
 - (ii) The Contractor shall, if called upon to do so, obtain the Inspector's approval of the way the Contractor proposes to supply the valves, penstocks and stoplogs and shall furnish such technical drawings, relevant documents and information as the Inspector may require.
 - (iii) The Contractor shall notify the Inspector in advance of the date on which any of the component parts, valves, penstocks or stoplogs will be ready for inspection.
 - (iv) The Inspector shall not be required to sign any form of waiver or indemnity concerning his presence or actions at the place of inspection.
 - (v) If any of the items, whether completed or in course of production, are rejected by the Inspector, they shall be marked or segregated in such manner satisfactory to the Inspector as to ensure their subsequent identification as rejected work.

- (vi) When independent tests and analyses, in addition to those made by the Inspector at the Contractor's or manufacturer's premises, are considered necessary by the Inspector, such tests or analyses will be made by persons appointed by the Inspector. The Contractor shall bear the cost of supply, labour, transport, testing, equipment and materials and carriage of samples and in addition where the results of such tests or analyses show that the items are not in accordance with the Specification, the cost of such tests or analyses.
- (vii) The Contractor shall not deliver any of the valves, stoplogs and penstocks and related parts or report them as ready for delivery to site until the Inspectors shall have given consent.
- (viii) The inspection, examination or testing including the approval of the valves, stoplogs and penstocks by the S.O. or Inspectors shall not release the Contractor from any of his obligations and liabilities under the Contract.
- (c) The Contractor shall allow in his rates where applicable any related expenses for inspection and testing of valves, penstocks and stoplogs by the S.O and his representatives.

G4.1.7 Defective Valves, Penstocks and Stoplogs

- (a) After inspection, no portion of the valves, penstocks or stoplogs shall in any way be damaged, repaired or altered. The S.O. or the Inspectors may reject any valve, penstock or stoplog which has been so treated, notwithstanding that it may have been previously passed.

G4.1.8 Rejected Valves, Penstocks and Stoplogs

- (a) Any valve, penstock or stoplog delivered to the Site which have been rejected by the S.O. or his Representative shall immediately be removed from the Site.
- (b) Any valve, penstock or stoplog which have been rejected shall be marked in a distinctive manner, which shall preclude any possibility of their use for the purposes of the Contract. Such valves/penstocks/stoplogs may be submitted for retest following the correction of any defect, where such corrections are permitted by S.O..

G4.1.9 Notices of Deliveries

- (a) The Contractor shall send to the S.O. advance notices of all consignments. Every consignment shall be accompanied by a detailed delivery note with the item number of each valve/stoplog and penstock.

G4.1.10 Marking and Packing

- (a) Each valve, penstock or stoplog shall be indelibly marked with the diameter and pressure rating and shall in addition carry a unique reference number to enable each item to be clearly identified with works fabrication records, works test certificates, delivery notes and the like. Wherever possible, the identification marks shall be painted on the outside of the item but where there is insufficient smooth surface area to accommodate the identification marks they shall be put on rust-proofed metal tags secured to the item with galvanised wire or chain (not through flange holes). The project equipment tag nos. shall also be included on each valve delivered to site.

- (b) Valves, penstocks and stoplogs shall be packed in the 'closed' position except for uncrated resilient seat gate valves which shall be in the 'open' position.

G4.1.11 Handling

- (a) The Contractor shall provide equipment as necessary to handle and install valves, penstocks and stoplogs and associated equipment without damage to exterior or interior coatings. The equipment shall include lifting beams, reinforced canvas slings, protective padding, cradles and the like. Wire rope or chain slings shall not be used for handling these items.
- (b) Temporary packing, coverings or crates provided for protection in transit shall not be removed (except for inspection purposes after which they shall be replaced) until immediately before installation.

G4.1.12 Protection against Damage during Transit

- (a) All valves, penstocks and stoplogs shall be securely packed in crates or boxes for protection against damage during transit. Valves of 450 mm diameter and above shall be individually crated, battened and bound with steel strip.
- (b) Flanges of valves shall be protected by wooden discs temporarily bolted on or secured by steel strapping. Spindle caps of sluice valves shall be removed and secured to the inner side of one of the wooden discs by means of a steel strip. Otherwise they shall be packed in a case. The projecting end of the spindle shall be well wrapped with straw rope covered by hessian cloth and secured in place by binding wire which shall be carried under the flange of the gland.
- (c) All plain ends shall be adequately protected by straw rope secured in place by binding wire or strap. None of the packings will be returnable. The cost of packing shall be included for in the Contract rates.

G4.1.13 Identification Plates and Labels

- (a) Each main and auxiliary item of the valves and penstocks shall, unless otherwise specified, have permanently attached to it in a conspicuous position an identification plate of weather resistant material on which are engraved or stamped the manufacturer's name, service rating, loading and duty at which the item of valves and penstocks has been designed to operate, serial number, weight and reference number.
- (b) Additionally, each item shall have marked upon its body in raised letters the manufacturer's name, service rating, year of manufacture and size of the valve and penstock.
- (c) Details of proposed inscriptions shall be submitted to S.O. for approval before any labels are manufactured.

G4.1.14 Installation of Valves

- (a) Valves shall be installed and commissioned in accordance with the manufacturer's instructions. After installation, valves shall be cleaned, gates, discs, seats and other moving parts closely inspected, foreign matter removed, and the valves checked for ease of operation. Moving parts shall be lightly greased or otherwise treated in accordance with the manufacturer's recommendations.

- (b) Except when otherwise specified or directed by the S.O., butterfly valves shall be enclosed in chambers and fixed with the disc spindle horizontal and supported as detailed on the Drawings, and installed so that when the valve is opening the lower portion of the disc moves in the direction of the main or normal flow.
- (c) Except where shown otherwise on the Drawings or agreed with the S.O., gate valves shall be fixed with their spindles vertical.
- (d) Valves without external gearing, and not otherwise required to be in a chamber, may be buried. The buried part of the valve shall be protected as specified. Except where shown otherwise on the Drawings, the valve shall be backfilled to just below the top of the valve or spindle shroud, and a surface box and tee - lug shall be provided.
- (e) Sufficient and adequate tie-rods, flange adapters and flange couplings shall be provided to ensure the valve can be easily dismantled and removed without dismantling any pipework or supports.
- (f) All dismantling joints shall be tied through suitable welded flange or brackets directly to the pipework such that hydraulic thrust will not be transmitted to pipework supports.
- (g) Jointing, sleeving, external wrapping, anchor and thrust blocks, valve chambers, valve marker posts and the cleaning and disinfection of valves shall be executed as specified for the associated pipeline.
- (h) Suitable flange adaptors shall be provided in the pipework adjacent to each valve of size DN 300 mm and above, unless otherwise specified.
- (i) Flange adaptors shall be tied to pipework with appropriate length of rods at each quadrant of pipe and not to pipework supports.
- (j) An access platform shall be provided for each valve on which the actuator is more than 1 m above the finished floor elevation with an access platform, safety rail and ladder according to the General Specification on Metalwork.
- (k) The Contractor shall design and provide all supports for valves. All valves shall be supported separately regardless of pipe supports. Supports shall be endorsed by P.E. and submitted the detail to S.O. for approval.
- (l) Where appropriate, ensure that open mesh or chequer plate or other flooring fits snugly around valve and penstock supports and are properly supported and sealed.
- (m) Unless otherwise specified, a corrosion allowance 0.01 mm per year, for 100-year design life, shall be provided for all valve internal wetted parts and external components.

G4.1.15 Works Test

- (a) Unless otherwise specified, all valves shall be tested fully according to the requirement in this specification as minimum.

- (b) After completion of assembly each valve shall be shop operated three times from the fully open position to the fully closed position and return to fully open under no flow condition to demonstrate that the assembly is working smoothly. The torque required on each valve shall be ascertained. Where electric motor actuators are provided the complete assembly of valve and actuator shall be tested.
- (c) All valves shall be hydrostatically tested at the place of manufacture to the pressures specified and shall satisfactorily pass the specified tests before they are packed for delivery.
- (d) All valves shall be body tested to 1.5 times the valve rating pressure. Seat tests to 1.1 times the valve allowable differential pressure shall be carried out on all valves.
- (e) All valves shall be tested in accordance with BS EN 12266 Part 1 and each valve shall be subjected to two separate hydrostatic tests as follows:
 - (i) Seat tests: The tightness of seats shall be tested as follows:
 - 1) With the wedge or disc closed and with the valve fixed at one end only and the other end free, the test pressure shall be applied to that face of the wedge or disc, the other face being at atmospheric pressure. There shall be no visible leakage past the wedge or disc at the hydrostatic test pressure (gauge) specified; and
 - 2) The above procedure shall be repeated but with the valve fixed at the other end and with the pressure applied to that end of the valve.
 - (ii) Body test
 - 1) With the wedge or disc open the test pressure (gauge) specified shall be applied to the whole body of the valve. There shall be no visible leakage through the metal, the flanged joint or the valve packing gland nor shall any part be permanently deformed.
 - 2) The test duration for all tests shall be as per the table below:

Nominal Diameter (mm)	Minimum test duration (minutes) for	
	Body	Seat
Up to and including 150	1	1
200 to 300	3	2
350 and above	5	3

G4.1.16 Site Test

- (a) Contractor shall be responsible for hydrostatic testing of the valves at site before actual process operation and S.O. shall to be notified prior to the test.
- (b) Contractor shall provide all necessary equipment and fittings and utilities for carrying out the site test. If the valves fail the site test, the Contractor shall either replace or rectify the defective valves at his own expense.

G4.1.17 Gate Valves

- (a) Standard gate valves up to and including 1000 mm nominal diameter shall conform to the requirements of BS EN 1171 for copper alloy faced or resilient seated valves with solid or split wedge, whilst gate valves up to and including 600 mm nominal diameter may alternatively conform to BS 5163-1, BS 5163-2, BS EN 1074-1 and BS EN 1074-2.
- (b) Gate valves larger than 1000 mm nominal diameter shall comply with the materials and other relevant provisions of BS EN 1171.
- (c) Gate valves up to 300 mm in diameter shall be of the resilient seal type. Gate valves of 350 mm in diameter and over shall be of the gunmetal faced seal type.
- (d) Valve spindles shall be of the rising type with see-through protective cover. There shall be provisions for special situations where non-rising spindle may be required subject to approval by the S.O.. The valve seal shall be replaceable with the valve fully open and the main under pressure.
- (e) Resilient seated valves shall have straight-through bores.
- (f) Unless otherwise specified, gate valves in chambers and other similar locations shall be provided with extension spindles with intermediate guides where necessary and hand wheels. Valves which are to be buried in the ground shall be provided with extension spindles, spindle caps, spindle supports, protection tubes and surface boxes as specified.
- (g) Gate valves shall be tested to BS EN 12266 Part 1.
- (h) Unless otherwise specified, gate valves shall be provided with indicators to show percentage (%) of opening and provision to facilitate locking at any random position over the entire operating range.

G4.1.18 Knife Gate Valve

- (a) General
 - (i) The knife gate valves, actuators and accessories shall be designed to fit into the spaces as specified in the specification. The facility structural and mechanical design shall not require modification to accommodate the knife gate valves.
 - (ii) Valves shall conform to AWWA C520-14 and be designed for specified application.
 - (iii) For Large knife gate valve of 800 mm diameter and greater, valve stroke time shall be adjustable between 400 to 900 seconds from fully open to fully closed for all hydraulically actuated valves. The valve stroke rate shall be 150 to 300 mm per minute for all electric motor actuated valves.
 - (iv) The manufacturer of the knife gate valves shall be ISO 9001 certified.
 - (v) Valves shall be of the fabricated bonneted type incorporating a round or square bottom gate.
 - (vi) Valves shall be designed for installation in a horizontal pipe with the bonnet in a vertical orientation unless otherwise specified.

- (vii) Valve design shall allow for the individual removal of all major components, such as the gate, bonnet, stem, yoke, and actuator without removing the valve body from the pipeline.
- (viii) All major components such as the valve body, gate, bonnet, and yoke are to have lifting lugs suitable to allow rigging for installation and/or maintenance removal if required.
- (b) Body
 - (i) The valve body shall be made manufactured from Type 316L stainless steel or cast ductile iron.
 - (ii) Body mating flanges shall be full round port flanges drilled to BS EN 1092 and of the PN rating of at least 1.5 times the maximum operating pressure of the piping system with machined raised gasket faces to MSS SP-6. Flanges shall be drilled for through-bolting except at the valve chest where drilled and tapped bolt holes shall be provided.
 - (iii) Valve body may incorporate hard-faced centreline buttons and nitronic chest buttons to achieve optimum shutoff at low differential pressures.
 - (iv) Knife gate valves shall incorporate flushing ports when used on process fluids containing settleable solids, including sludges and used water. Sufficient flushing ports shall be provided to clear any debris which could impede the operation of the valve. As a minimum two flushing ports shall be provide to flush the valve seat. Inside diameter of the valve body shall be equal to or greater than the adjoining pipe inside diameter. Piping inside diameters will be equivalent to the valve diameters.
- (c) Seats
 - (i) Valve seats shall incorporate a bearing surface and a sealing surface for tight shutoff in one direction. The seat design shall be full port or reduced port with the flow area no less than 90 percent of nominal valve size area.
 - (ii) The sealing surface shall be a suitable synthetic material which has been proven over time in other valve designs.
 - (iii) The bearing surface shall be Ultra High Molecular Weight (UHMW) polyethylene.
 - (iv) The seat design shall incorporate gate guides and seating wedges adequate to withstand full reverse pressure without damage with the gate in the fully CLOSED position.
 - (v) Valve seat shall be replaceable without removing valve body from pipeline.
- (d) Gate
 - (i) Fabricate from Type 316L stainless steel plate or casting. Contacting surfaces shall be finished to 2 microns rms or better. The gate shall be solid stainless steel and not include ribs, gussets, etc. for reinforcing. The gate shall have a bevel knife-edge for cutting through solids.

- (e) Stem
 - (i) The valve stem shall be Type 316L stainless steel. Valve stem must be detachable from the knife gate assembly for removal without removing the valve bonnet or stem. It shall consist of two pieces, a non-threaded portion inside the bonnet and a threaded portion detachable without removing the valve bonnet.
- (f) Rising Stem Type
 - (i) Contact surfaces of stem threads shall be rolled or machined and polished to a 2 microns rms finish or better. Stem threads are to be left hand ACME designed for optimal strength and to allow the valve operating speeds to be met; and
 - (ii) Provide with stem covers and rod boots.
- (g) Bonnet
 - (i) The valve bonnet shall be fully rated to the field pressure and manufactured from Type 316L stainless steel or cast ductile iron.
 - (ii) The valve bonnet shall be a stem packed design with multiple layers of braided asbestos free Teflon impregnated packing or self-adjusting chevron type packing rings of a suitable material to prevent atmospheric leakage. The bonnet shall be complete with a packing gland.
 - (iii) The design of the bonnet shall incorporate features that provide access to the interior of the valve for maintenance, inspection, assembly and disassembly.
 - (iv) The bonnet shall include adequate venting and flushing ports.
 - (v) Prior to machining, all pressure boundary welds shall be liquid penetrant inspected by a (ASNT) Level II certified (by test) inspector. All other non-pressure boundary welds shall be inspected by (AWS) QC-1 certified welding inspector. A certificate of compliance shall be submitted.
- (h) Yoke
 - (i) The valve yoke shall be fabricated from Type 316L stainless steel and designed to support the hydraulic or electric valve operator.
 - (ii) The yoke, on electric operated valves, shall be extended such that the electric actuator is 1.0 metre above the operating level finished floor elevation.
- (i) Flange Bolting and Gaskets
 - (i) Provide bolt, nuts, washers and gaskets required for mating one flange of each valve. Bolts, nuts and washers shall be A4-70 material as per BS EN ISO 3506-2. All bolts, nuts, washers and gaskets shall be suitable for the valve flanges. Anti-seize compound shall be applied to prevent seizure and galling of the fasteners. Gaskets shall be of a suitable material and 3 mm minimum thickness.
- (j) Valve Fasteners
 - (i) Provide metric fasteners constructed of a material with a higher galling limit than 316L stainless steel of equal or higher corrosion resistance than 316L stainless steel.

G4.1.19 Large Bore Knife Gate Valve

- (a) Large bore knife gate valves shall be those that are of nominal diameter sizes of 650 mm or larger.
- (b) A single valve manufacturer shall be made responsible for supplying the large bore knife gate valves and for co-ordination of design, assembly, testing, and installation of each valve; however, this does not relieve the Contractor of the responsibility to comply with the requirements in this Section. Each valve shall be produced and assembled by the manufacturer at a facility owned or operated by the manufacturer and under the direct supervision and control of the manufacturer.
- (c) The manufacturer of the knife gate valves shall have at least ten (10) years' experience in the design, fabrication, and supply of fabricated bonneted knife gate valves. This experience shall include the design, fabrication, and supply of at least one knife gate valve 2,400 mm in diameter or larger. Valve or valves shall have been in actual operational service for a minimum of 3 years.
- (d) For each valve size and pressure rating, there shall be a finite element analysis (FEA) of the complete valve assembly and individual components such as valve body, and gate. The maximum allowable stresses shall not exceed 80 percent yield strength of the material of construction. The FEA shall be conducted at the body shell test pressure and at the maximum seat test pressure for the body and gate combination with the gate in the closed position. Material yield strength shall be calculated using the 0.20 percent offset method based on test data of material at 25 deg. C.
- (e) The valves shall be designed to fit into the spaces provided as shown on the 3D Model. The facility structural and mechanical design shall not require modification to accommodate the valves.
- (f) Valves shall be designed for use in water applications. All wetted and non-wetted parts, including fasteners shall be constructed of Type 316 stainless steel for optimal corrosion resistance.
- (g) The manufacturer of the knife gate valves shall be ISO 9001 certified.
- (h) The valve construction configuration shall be as follows:
 - (i) Valves shall be of the fabricated bonneted type incorporating a round or square bottom gate.
 - (ii) Valve components, such as seats, stems, bonnets, stems, yokes, discs and actuators are to be replaceable without removing the valve from the pipeline.
 - (iii) Valves shall be designed for installation in a horizontal pipe with the bonnet in a vertical orientation.
 - (iv) Valve design shall allow for the removal of the gate without removing the valve body from the pipeline.
 - (v) All major components such as the valve body and gate are to have lifting lugs suitable to allow rigging for installation and gate removal.
 - (vi) The valve body shall be made with fabricated Type 316L stainless steel flanges and gaskets.

- (vii) Body mating flanges shall be full round port flanges drilled to BS EN 1092 PN16 standard with machined raised gasket faces to MSS SP-6. Flanges shall be drilled for through-bolting except at the valve chest where drilled and tapped bolt holes shall be provided.
- (viii) Valve body may incorporate hard-faced centreline buttons and nitronic chest buttons to achieve optimum shutoff at low differential pressures.
- (ix) The inside diameter of the valve body shall be equal to or greater than the adjoining pipe inside diameter.
- (x) Valve seats shall incorporate a bearing surface and a sealing surface for tight shutoff. The seat design shall be full port or reduced port with the flow area no less than 90 percent of nominal valve size area.
- (xi) The sealing surface shall be a suitable synthetic material which has been proven over time in other valve designs.
- (xii) The bearing surface shall be Ultra High Molecular Weight (UHMW) polyethylene.
- (xiii) Valve gate shall be fabricated from Type 316L stainless steel plate or casting. Contacting surfaces shall be finished to 2 microns rms or better. The gate shall be solid stainless steel and not include ribs, gussets, etc., for reinforcing. The gate shall have a bevel knife-edge for cutting through solids.
- (i) Valve body shall incorporate a sufficient quantity of flushing ports to allow for the flushing of the valve interior should it be necessary. At a minimum, two 75 mm seat flushing ports shall be provided to flush each valve seat. These flushing ports shall be provided with non-lubricated type, flanged eccentric plug valves for isolation.
- (j) The valve bonnet shall be fully rated to the field pressure and fabricated from Type 316L stainless steel bonnet flanges and gussets.
- (k) The valve bonnet shall be a stem packed design with multiple layers of braided asbestos free Teflon impregnated packing or self adjusting chevron type packing rings of a suitable material to prevent atmospheric leakage.
- (l) The bonnet shall be complete with a fabricated Type 316L stainless steel packing gland. The design of the bonnet shall incorporate features that provide access to the interior of the valve for maintenance, inspection, assembly and disassembly.
- (m) The bonnet shall include adequate venting and flushing ports.
- (n) Each valve shall be provided with integrated bypass piping with bypass valve.
- (o) Provide metric fasteners constructed of a material with a higher galling limit and corrosion resistance than Type 316L stainless steel.
- (p) Provide bolt, nuts, washers and gaskets required for mating one flange of each valve. Bolts, nuts and washers shall be A4-70 material as per BS EN ISO 3506. All bolts, nuts, washers and gaskets shall be suitable for BS EN PN 1092 PN16 flanges.
- (q) Prior to thermal stress relieving and machining, all pressure boundary welds shall be liquid penetrant inspected by a (ASNT) Level II certified (by test) inspector.

- (r) All non-pressure boundary welds shall be inspected by (AWS) QC-1 certified welding inspector. A certificate of compliance shall be supplied.
- (s) A corrosion allowance 0.01 mm per year, for a 100-year duration shall be provided for all valve internal wetted parts and external components.
- (t) Each valve supplied shall be subjected to the following tests:
 - (i) Body Shell Test as per BS EN 12266-1 or ANSI/AWWA C520-14; and
 - (ii) Seat Test as per BS EN 12266-1 or ANSI/AWWA C520-14.
- (u) The body shell and seat testing of the knife gate valves shall be conducted and completed within a 150-day time period after the S.O. is notified of the test facility availability.
- (v) If any of the knife gate valves do not meet the specified minimum performance requirements criteria during the initial performance tests, the Contractor will have 45 days in which to modify the equipment and/or operation and to request a retest, completely at the Contractor's expense. The Contractor will be permitted up to two retests for each knife gate. If the knife gate valves cannot meet the performance requirements after the two retests, the knife gates will be considered defective. Such valves shall be replaced with all incurred costs to be borne by the Contractor.
- (w) Factory trained engineers of the knife gate valve manufacturer, who have complete knowledge and experience in the proper installation, commissioning and operation of the equipment, shall inspect the final installation and assist the Contractor with all site functional tests of the equipment.
- (x) Functional testing shall be conducted after the installation of the knife gate valves. Each complete unit shall be subject to field functional tests under actual, or simulated, operating conditions to determine that operation is satisfactory and in compliance with the Specifications.
- (y) The Contractor shall provide, calibrate, and install all temporary equipment required for the functional tests. Written test procedures will be submitted to the S.O. for approval at least 30 days prior to testing.
- (z) Under all operational conditions, the functional tests shall demonstrate that the valves shall:
 - (i) Have not been damaged by transportation or installation;
 - (ii) Have been properly installed;
 - (iii) Have no mechanical defects;
 - (iv) Are properly aligned; and
 - (v) Have been properly connected.

- (aa) Technical specification requirements to be complied with for each of the large bore knife gate valves shall be as follows:

Requirements for Large Bore Knife Gate Valves

No.	Specification/Parameter Description	Requirement	Remark
1.0	Type	Electrically actuated metal seated knife gate	
2.0	Design Code/Standard	AWWA C520-14 Knife Gate Valves, 2 In. (50 mm) through 96 In. (2,400 mm)	
3.0	Pressure rating	1.5 times pump shutoff head or 1.5 times maximum operating pressure, whichever is higher	
4.0	Ends Connection	PN rated raised-face flange to BS EN 1092 or Class rated flat-face flange to ASME B16.1	Preference for BS EN 1092. Contractor to ensure compatibility of connecting pipe flanges. Flanges flatness shall be machined parallel to each other to within 0.005 inch.
5.0	Valve position upon power failure	Fail to Close	
6.0	Major valve components	Body, gate and operating unit (actuator)	
7.0	Materials of Construction		
7.1	Body	Stainless steel ASTM A240 Type 316L	
7.2	Bonnet	Stainless steel ASTM A240 Type 316L	
7.3	Knife plate	Stainless steel ASTM A240 Type 316L	
7.4	Seat ring	Stainless steel ASTM A240 Type 316L	
7.5	Gland	Stainless steel ASTM A240 Type 316L	
7.6	Gland packing	Teflon impregnated non-asbestos + rubber	
7.7	Yoke	Stainless steel ASTM A240 Type 316L	(where applicable)
7.8	Stem	Stainless steel ASTM A479 Type 316L	

No.	Specification/Parameter Description	Requirement	Remark
7.9	Gearbox	Manufacturer's standard with compliance to General Specification	
7.10	Electrical Actuator	Manufacturer's standard with compliance to General Specification	
8.0	Appurtenances		
8.1	Drain opening	Required drilled and tapped drain opening provided with plug on valve body.	(where applicable)
8.2	Vent opening	Required drilled and tapped vent opening provided with plug on valve body	(where applicable)

G4.1.20 Butterfly valves

- (a) The shaft shall be fabricated in stainless steel if it is exposed to the pipeline fluid.
- (b) Butterfly valves shall comply with BS EN 593:2009+A1 except as specified herein. Valves shall be suitable for mounting in any position.
- (c) The Contractor shall provide details of the materials of manufacture and the design of butterfly valves, including the access route to repair or replace seals. Evidence shall be provided to show that the proposed materials and designs for the sealing and seating arrangements have given satisfactory performance in similar conditions elsewhere. Valves shall have compliant Proof of Design Tests as specified by section 5.1.4 of AWWA C 504.
- (d) In general, eccentric type butterfly valves shall be selected for used water applications whereas concentric type butterfly valves shall be selected for clean water applications unless otherwise specified. The selection shall take account of compatibility with the process fluid, corrosion resistance durability, pressure rating and leak tightness.
- (e) Flanges and hubs for shaft bearing housings shall be integrally cast with the valve body except where otherwise specified. All bearings shall be of a type, which is independent of external lubrication.
- (f) Hollow discs, and any other castings with voids, shall be fitted with threaded plugs.
- (g) Discs shall be capable of maintaining their fully open and fully closed positions with the gearbox removed. Limit stops shall be capable of absorbing full operating torque with a minimum design safety factor of 5.
- (h) The shaft and disc fixing shall be capable of absorbing the full operating torque with a minimum design safety factor of 5. Shaft seals, when used, shall be of the resilient O-ring type. Packing shall be either of the resilient O-ring or self-adjusting Chevron type.

- (i) Unless otherwise specified, valves shall comply the following as a minimum:
 - (i) Flanged or wafer type short body type;
 - (ii) Seat shall be replaceable without removing the valve from the pipeline;
 - (iii) Bubble-tight with rated pressure applied from either side;
 - (iv) No travel stops for the disc on interior of the body;
 - (v) Self-adjusting V type or O ring shaft seals;
 - (vi) With isolation of metal-to-metal thrust bearing surfaces from flow stream;
 - (vii) Non-metallic or coated metallic permanently lubricated bearings; and
 - (viii) With valve position indicator.
- (j) Gate disc shall rotate 90 degrees from the full open to tight shut position.
- (k) When subjected to the maximum design head, a stress safety factor of 3 on the yield point or 5 on the ultimate strength whichever is lower shall be considered.
- (l) Maximum deflection of the obturator structural design limit shall be 0.8 mm.
- (m) Experimental units or developmental designs will not be considered.
- (n) Butterfly valves designed to be clamped between two pipework flanges shall have sealing arrangement compatible with the mating pipework flanges. Adaptor pieces where needed to ensure correct mating with flange adaptors shall be subject to the consent of the S.O..
- (o) Unless otherwise specified, the butterfly valve shall be accordance with BS EN 1563 or equivalent, with ductile iron body and stainless steel 316 disc, stainless steel shaft to BS EN 10277-1, BS EN 10277-2, BS EN 10277-3, BS EN 10277-4, BS EN 10277-5 and BS EN 10278 or equivalent, EPDM with WRC (Water Research Commission) approval in accordance with BS 6920-1, BS 6920-2 and BS 6920-3 liner, PN10 or PN16 as specified and zero leakage.
- (p) Butterfly valves shall be arranged with horizontal operating spindles unless otherwise specified or approved by S.O..

G4.1.21 Resilient Seated Valves

- (a) In the event resilient seated been specified, the valve seats shall be resilient, durable, and replaceable and formed of approved material. The valve seat shall be securely clamped either into a machined groove in the valve body or to the edge of the disc by seat retention members or other equivalent retention device, in such a manner as to prevent leakage of water under the seat and to hold the seat securely in position during opening and closing of the valve disc. The seat retention members shall be of stainless steel and shall be securely fastened to the body or disc with stainless steel fasteners.
- (b) The seat and its retention members shall be adjustable to ensure water tightness and to minimise the seating and unseating friction forces. When all the seat retention members are in place, the finished edges of the seat shall fit closely and the surface shall be smooth with all fastenings set flush in the water passage to offer the least resistance possible to the flow of water through the valve.

- (c) Valve seats which extend over the face of the flanges to secure the seat in place, or which require surface grinding and/or hand fitting of the disc, or designs which require the adjoining pipe flange to retain the seat in place and resist line pressure, are not acceptable.

G4.1.22 Metal Seated Valves

- (a) In the event metal seated been specified, the valves shall have metal to metal seating, and shall be of the low leakage or regulating type or both, as particularly specified.
- (b) Regulating valves shall be designed for long-period operation in the partly closed, throttled position.

G4.1.23 Air Release Valves

- (a) Air release valves shall be provided for installation at high points or changes in gradient in pipelines or at specified locations, to fulfil the following functions:
 - (i) To exhaust air from a pipeline automatically when it is being filled with water at a sufficiently high rate to prevent back pressure restriction of water inflow rate;
 - (ii) To ventilate a pipeline automatically when it is being emptied of water at a sufficiently high air inflow rate to prevent the development of vacuum conditions in the pipelines; and
 - (iii) To release air accumulating in pipelines automatically during normal working conditions.
- (b) Unless otherwise specified, air valves shall be of one of the following types:
 - (i) Double orifice type with a large orifice for automatic ventilation or exhaust of the pipeline and a smaller orifice for automatic release of air under normal working pressure;
 - (ii) Single large orifice type for automatic ventilation or exhaust of pipeline;
 - (iii) Single small orifice type for automatic release of air under normal working pressure; and
 - (iv) The valve shall be equipped with an integral surge alleviation mechanism that automatically dampens surge pressures due to rapid air discharge or the subsequent re-joining of separated water columns.
- (c) Large orifice air valves, including those incorporated in double air valves, shall be constructed so that the air flow actively holds the valve open during the discharge of air at all flows up to and including sonic velocities. When air valves are coupled to their respective isolating valves, they shall be capable of admitting to or exhausting from the pipeline to which they are connected the required quantities of free air without the pressure differential across the combined air valve and isolating valve exceeding 0.5 bar unless otherwise specified or approved by S.O..
- (d) The ball of the large orifice shall bear a calculated mathematical relation with the inlet diameter of the valve i.e. average cross-sectional area of escaping air stream, so that the ball will be blown shut by a stream of water but held down by a stream of air. The buoyancy of the ball shall be such that it will ensure effective sealing of the large orifice even at low pressure. The weights of each ball of the same size and type shall not differ by more than 2 %.

- (e) Small orifice air valves, including those incorporated in double air valves, shall be capable of opening and discharging up to not less than 0.5 m³/min of free air when the pressure in the pipeline is at the maximum working pressure for which the valve is designed.
- (f) All valve and flanged shall be drilled to BS EN 1092-2:1997 and of the PN rating that is at least 1.5 times the maximum operating pressure of the main pipe.
- (g) All valves shall be reliable in action and shall operate such that the ball cannot be held against orifice by air pressure alone.
- (h) Balls or floats shall be of ABS plastic, vulcanite, rubber-covered metal or stainless steel, and shall operate automatically at all pressures up to the test pressure when installed in their valves.
- (i) Orifices shall be bronze or stainless steel no less than 3 mm and tapering to 10 mm suitable to release accumulated air within the pipe. The profile of the orifice shall be such as to avoid damage to the ball surface. The orifice shall be protected by a suitable plug of stainless steel.
- (j) Air valves shall be designed so that each float seats against its orifice or causes the orifice to be closed without leakage. Balls and seats shall be designed to minimise the risk of adhesion of the ball to the seat. The valve type shall be proven by experience to be suitable for the specified duties. Under no circumstances shall the large orifice ball blow shut prematurely.
- (k) Air valve specified for sewage containing solid and fibrous matter shall be designed such that efficient release of air is not hindered by solid matter in the sewage.
- (l) Each air valve shall be provided with an isolating valve. Air valves with a connection not exceeding 25 mm diameter to the main pipeline shall each have an integral lever-operated isolating ball valve. All other air valves shall have a separate resilient seated, double flanged isolating butterfly valve with lever operation.
- (m) All air valves shall be made insect proof by providing stainless steel screens at the vents leading into the atmosphere.
- (n) Valve shall be cast iron casing as minimum unless otherwise specified.
- (o) Each small orifice or double air valve shall be fitted with a test cock in the valve body to permit easy verification that the small orifice air valve is operating properly and that its orifice is not blocked.
- (p) The valve body assembly shall be hydraulically tested to one and a half (1.5) times the valve rating pressure for a minimum period of 5 minutes and thereafter compressed air tested at a slightly higher pressure than that at which the air shall be released through the valve inlet to check the function of the ball. Subsequently, the hydraulic pressure shall be reduced to 0.2 bar or specified rating which agreed by S.O. and there shall be no leakage through any of the orifices. The test shall be carried out in the presence of the S.O. or his representative as witness and the certified test result by the manufacturer shall be submitted to S.O. for approval.
- (q) During maximum rated air flows, the pressure differential across the valves shall not exceed 1.4 bar.

- (r) Curves for free air discharges and inflow against various air pressures at valve inlet shall be submitted to S.O. for approval. These curves must be endorsed and signed by both the manufacturer and agent/sole distributor, without which their valves will not be accepted. In this instance, an endorsement means that the manufacturer and agent/sole distributor warrants the results of the tests and that each of the valves manufactured will meet these standards as a minimum.

G4.1.24 Combination Air Valves

- (a) Each combination air valve shall serve the following functions:
 - (i) Exhaust large quantities of air when the concerned piping system is being filled and the valve shall close upon liquid entry;
 - (ii) Open to admit air into the concerned piping system when the piping system is draining or when a negative pressure occurs; and
 - (iii) Release accumulated air from the concerned piping system while the system is in operation and under pressure.
- (b) Valves shall be manufactured and tested in accordance with American Water Works Association (AWWA) Standard C512.
- (c) AWWA M51 Air Valves: Air-Release, Air/Vacuum and Combination shall be referred for guidelines for selecting, sizing, locating and installing the combination air valves.
- (d) The manufacturer shall demonstrate a minimum of (5) years' experience in the manufacture of wastewater air valves.
- (e) Combination air valve sizing calculations for valve diameter and orifice openings shall be submitted for approval by the S.O..
- (f) The valve shall either be in a single body or dual body configurations.
- (g) The connections associated with each combination air valve shall comply with the following requirements:
 - (i) Single body valves sizes 4 in. (100 mm) and smaller shall have full size NPT inlets and outlets equal to the nominal valve size;
 - (ii) The body inlet connections shall be hexagonal for a wrench connection;
 - (iii) The body shall have 2" NPT cleanout and 1" NPT drain connection on the side of the casting;
 - (iv) Dual body valves sizes 3 in. (75 mm) and smaller shall have full size NPT inlets and outlets equal to the nominal valve size. The body inlet connection shall be hexagonal for a wrench connection;
 - (v) Valve sizes 4 in. (100 mm) through 6 in. (150 mm) shall have bolted flanged inlets and NPT outlets;
 - (vi) Size 8 in. (200 mm) valves shall have flanged inlets and outlets. Flanges shall be in accordance with BS EN 1092 PN16 or ANSI B16.1 for Class 250 raised face flanges; and
 - (vii) The valve shall have three additional NPT connections for the addition of backwash accessories.

- (h) Both single and dual body valves shall provide an extended body with a through flow area equal to the nominal size.
- (i) Floats shall be unconditionally guaranteed against failure including pressure surges. A resilient bumper shall be provided on 4 in. (100 mm) and larger sizes to cushion the float during sudden opening conditions. The seat shall provide drop tight shut off to the full valve pressure rating.
- (j) Dual body valves shall consist of an Air Release Valve piped to an Air/Vacuum Valve with a full-ported stainless steel SS316 ball valve.
- (k) The Air Release Valve shall have an extended leverage mechanism with sufficient mechanical advantage so that the valve will open under full operating pressure to release air. An adjustable threaded resilient orifice button shall be used to seal the precision discharge orifice in the cover. Air gaps shall retard waste solids from clogging the upper shut-off float of the large orifice and the lever mechanism of the small orifice valve.
- (l) The large orifice shall enable the venting of large volumes of air whilst the small orifice enables the venting of small air pockets. Should vacuum develop in the piping force main, both orifices shall open to allow air to re-enter to break the vacuum.
- (m) The Air/Vacuum Valve sizes 4 in. (100 mm) and larger shall have a cover fitted to the valve body by means of a machined register to maintain concentricity between the top and bottom guide bushings at all times. The tandem float assembly shall have a hexagonal guide shaft supported in the body by circular bushings to prevent binding from debris. The upper float shall be protected against direct water impact by an internal baffle. The seat shall be a minimum of 0.5 in. (12 mm) thick on 2 in. (50 mm) and larger valves and secured in such a manner as to prevent distortion.
- (n) Single body valves shall have a full port orifice, a double guided plug, and an adjustable threaded orifice button. Globe style body design shall be employed as necessary to increase float clearance and reduce clogging. The plug shall be protected against direct water impact by an internal baffle and an extended float stem. The plug shall have a precision orifice drilled through the center stem. The float shall include a sensitivity skirt to minimize spillage.
- (o) The following accessories shall be provided with each combination air valve:
 - (i) A Regulated Exhaust Device shall be provided to reduce pressure surges due to column separation or rapid changes in velocity and pressure in the pipeline;
 - (ii) The Regulated Exhaust Device shall be mounted on the outlet of the Combination Air Valve to allow free air flow in and out of the valve, close upon rapid air exhaust, and control the air exhaust rate to reduce pressure surges;
 - (iii) The device shall have a threaded or flanged globe-style body with a center guided disc and seat assembly. The disc shall have threaded holes to provide adjustment of the air exhaust rate through the valve. The holes shall provide for a flow area of 5% of the nominal valve size;
 - (iv) The material of the body shall be consistent with the Combination Air Valve. The seat and disc shall be ASTM A351 Grade CF8M stainless steel; and

- (v) Backwash accessories shall be furnished when specified and shall consist of an inlet shut-off valve, a blow-off valve, a clean water inlet valve, rubber supply hose, and quick disconnect couplings. Accessory valves shall be quarter-turn, full ported stainless steel SS316 ball valves.
- (p) The exhaust ports of each combination air valve shall be piped to an approved location as shown in the 3D model or as agreed by the S.O. where required to be site routed during works execution for safe discharge without causing noise, nuisance sprays, spills or jets to the surrounding.
- (q) The valves shall be installed such that there shall be no obstruction to their inlet ports for air admission.
- (r) The exterior of each valve shall be coated with a universal alkyd primer.
- (s) Each combination air valve shall comply with the requirements in the following tabulation:

Requirements for Combination Air Valves

No.	Specification/Parameter Description	Requirement	Remark
1.0	Type	Combination Air	
2.0	Design Code/Standard	ANSI/AWWA C512	
3.0	Pressure rating	PN16 according to BS EN 1333 or Class 250 according to ASME B16.1	
4.0	Ends Connection	Raised face flange to BS EN 1092 PN16 or Flat-face flange to ASME B16.1 Class 250	Preference for BS EN 1092. Contractor to ensure compatibility of connecting pipe flanges. Flanges flatness shall be machined parallel to each other to within 0.005 inch.
5.0	Materials of Construction		
5.1	Body	ASTM A126 Class B Cast Iron	
5.2	Cover	ASTM A126 Class B Cast Iron	
5.3	Float	Type 316 stainless steel	
5.4	Plug	Type 316 stainless steel	
5.5	Guide Shaft	Type 316 stainless steel	Non-metallic materials are unacceptable.
5.6	Bushing	Type 316 stainless steel	Non-metallic materials are unacceptable.
5.7	Seat	Buna-N	

No.	Specification/Parameter Description	Requirement	Remark
6.0	Appurtenances		
6.1	Regulated Exhaust Device	To be provided for each combination air valve	
6.2	Backwash accessories	(Refer to the descriptive specifications for details)	As per manufacturer's recommendations.

G4.1.25 Non-Return Valves

(a) General

- (i) Unless otherwise specified, all metallic Non-Return valves shall conform to BS EN 12334 and shall be double flanged, swing type, faced with copper alloy, nickel alloy or stainless steel. They shall have rapid closure characteristics with minimum shock on closing. Valves shall be tested to BS EN 12266 Part 1.
- (ii) Thermoplastic check valves shall conform to BS EN ISO 16137, BS ISO 9393 Parts 1 and 2, BS EN 917:1997, BS EN ISO 1167-1 and BS EN ISO 1167-2.
- (iii) Valve bodies shall be capable of withstanding 1.5 times the rated pressure.
- (iv) Valves shall have a pressure rating of at least 1.5 times the maximum operating piping system pressure and flanges shall be to BS EN 1092-2 of equivalent pressure rating, unless otherwise specified.
- (v) Material of construction shall be selected to avoid electrolytic corrosion and erosion.
- (vi) The internal and external surfaces of the valve body shall be manufactured from corrosion resistant material or be with corrosion resistant coating and shall be approved by S.O.;
- (vii) Flow velocity through the valve shall not exceed 2.5 m/s.
- (viii) All check valve shall be provided with a position indicator of the disc, visible from all positions.

(b) Non Return Valves (Nozzle type)

- (i) Nozzle type non-return valve shall be used where specified as necessary for the concerned applications. It shall be double flanged, nozzle type and designed in such a way that the valve disc will close rapidly by the assisted spring under all operating conditions without a slam. The moving masses of the disc shall be extremely small to ensure fast closing. The non-slam characteristic shall be attained by valve reacting quickly to flow changes so as to always outpace the process change. No external dampening devices shall be accepted on the valve.
- (ii) The valve size and design shall be selected to give the optimum performance in respect of its rapid closure taking full account of the system within which the valve is to be installed, especially when the system includes air vessels or when the system static head is high.

- (iii) The valve shall be designed for long service life and be free maintenance having no soft parts to wear, cause misalignment or need for replacement. The sealing shall be metal to metal and drop tight. It shall also have aerodynamic flow pattern design to ensure a very low-pressure loss.
 - (iv) Valves shall be suitable for use with clean water systems or systems containing only very small water borne particles.
 - (v) Nozzle check valves shall be of the disc or ring type sealing on to a metal seat by one or more guides and helical springs.
 - (vi) The valves shall be suitable for operation when installed in any position.
 - (vii) Rapid closure shall be ensured by keeping both the extent of movement and the inertia of the working component as small as possible without causing an unduly large head loss through the open valve.
 - (viii) Such valves may incorporate a number of relatively small water passages. Closure of the valve shall be by short closing stroke, and the closure time shall be controlled by the action of the pre-loaded spring.
 - (ix) Valves shall be supplied with works test certificates indicating head-loss across the valve and closure characteristics at specified flow.
 - (x) The valve body and diffuser shall be of grey iron GG25 to DIN1693 or ASTM A216WCB. The valve disc, spring, guide bush and seat ring shall be of stainless steel AISI 316 or ASTM CF8M. The spring material shall be stainless steel. The valves shall not contain any brasses which have more than 5% zinc. The interior of the casing and disc shall be coated with non-toxic paint to a minimum dry film thickness of 75 microns. Lifting lugs on the body shall be provided for ease of installation and removal.
 - (xi) All non-return valves shall be hydrostatically tested to BS EN 12266 part 1. Test certificates shall be forwarded to the S.O. before the valves are delivered site. The sub-contractor shall be responsible for testing the valves at site before installation, and if the valves fail at site test, the sub-contractor shall either replace or rectify the defective valves at their own expense. If the site test must be carried out in an open space, the sub-contractor shall erect a temporary shelter and provide proper lifting facilities to minimise the duration of site test. The test at site shall include the following:
 - 1) Tightness of seat at rated pressure (the maximum allowable leakage rate shall be $0.01 \text{ mm}^2/\text{s} \times \text{DN}$); and
 - 2) Tightness of body at $1 \frac{1}{2}$ times the rated pressure.
- (c) Non-Return Valves (Non-slamming)
- (i) Check valves shall comply with BS EN 12334 or the equivalent international standard and shall be of the swing type suitable for either vertical or horizontal use. With clean water application, Dual-flaps or split disc check valves may be used unless otherwise specified. Ball type Non-return valve shall be used if specified in the specification or drawings. For used water applications, swing check valves shall be installed with the pipeline axis horizontal unless otherwise approved by the S.O..

- (ii) The valve specified for non-slamming application shall ensure closure in the shortest possible time following deceleration of the water column, ideally reaching its seat without slamming at the instant forward motion of the column ceases.
 - (iii) Valve minimum material requirement for the bodies, discs and covers, shall be ductile iron BS EN 1563, Body and door seats shall be Gunmetal to BS EN 1982, for all seals shall be EPDM whilst spindles and hinge pins shall be stainless steel to BS EN 10088. Resilient seats shall be used for all wastewater application.
 - (iv) Except for wafer split disc check valves, valves shall be flanged to BS EN 1092, and rated to at least 1.5 times the maximum operating piping system pressure.
 - (v) With exception of Ball type non-return valves, valves shall be fitted with external levers and balance weights complete with guards.
 - (vi) With exception of Ball type non-return valves, valves shall be fitted with limit and proximity switches. A positive action of the switches shall be obtained within 10 degrees of the valve seating position. Limit switch housings shall have an Index of Protection rating IP 55. Limit and proximity switches shall be packed separately for delivery to prevent damage in transit.
 - (vii) Valve shall be resilient seated and suitable for the type of installation as specified. It shall be non-slam operation with by-pass bosses. The valve shall be with fusion bonded epoxy coating.
 - (viii) During normally open position, head loss must be at the minimum possible and in the close position, there shall be no leak.
 - (ix) Friction curves and dynamic performance test for non-slam features must be submitted to S.O. for approval. These curves must be endorsed and signed by both the manufacturer and supplier/distributor, without which their products will not be accepted. The endorsement means that the manufacturer and supplier/distributor certifies the results of the tests and that each of the valves manufactured will meet these minimum standards.
 - (x) Conventional non-return valves will not be accepted in this application.
 - (xi) Limited and proximity sensor shall be provided for the Recoil type non-slamming non-return valve.
- (d) Ball Check valve
- (i) Ball check valves shall conform to the requirements of BS EN 12334 and BS EN 12266-1 as far as applicable.
 - (ii) Ball check valves shall provide a positive back flow prevention, incur low friction loss in the forward flow, be able to be installed horizontally or vertically and be reliable with minimum maintenance.
 - (iii) There shall be 3 openings in ball check valve housing: one for fluid inflow, one for fluid outflow and another for positioning the ball inside the valve housing which is closed by a bolter cover.
 - (iv) Depending on the specific application, ball check valves may be provided with either a floating or sinking ball.

- (v) Ball check valves shall be capable of keeping solids, stringy material, grit, rags, etc. moving without the need for back flushing. The ball shall roll clear of the water flow path providing full flow equal to the connected nominal pipe size.
- (vi) The ball shall be lined with NBR, EPDM or alternative material for compatibility with the process fluid and corrosion resistance.
- (vii) Body of ball check valves shall be nodular cast iron type GGG 40/ASTM 65-45-12/SAE 4512 or better materials as may be required to be corrosion resistant and suit the process fluid operating conditions.

G4.1.26 Pressure and Flow Control Valve

- (a) Pressure control valves and flow control valves shall be designed for the operating conditions as specified.
- (b) The valves shall be capable of controlling the required parameter of flow or pressure within plus and minus 5 percent of the set value. The rate of response of opening and closing of the main valves shall be controllable at the valve. Means for external indication of the position of the valve element shall be fitted.
- (c) Flow control valves shall be designed to prevent the flow downstream rising above that specified in the specification, regardless of the operating pressures in the system upstream or downstream of the valve.
 - (i) Pressure reducing valves
 - 1) Pressure reducing valves shall be designed to reduce a constant or variable inlet pressure to a predetermined constant outlet pressure, at flows varying from the maximum capacity of the valve to zero flow. The outlet pressure shall be adjustable. A pressure gauge shall be provided to indicate downstream pressure over the operating range of the valve; and
 - 2) The operating discharge pressure range shall be between upstream pipeline pressure and zero.
 - (ii) Pressure sustaining valves
 - 1) Pressure sustaining valves shall be designed to maintain the pressure in the pipeline immediately upstream of the valve at or above a preset value, irrespective of the flow and pressure conditions downstream of the valve; and
 - 2) The upstream pressure shall be adjustable. A pressure gauge shall be provided to indicate upstream pressure over the operating range of the valve.
 - (iii) Pressure Relief Valves
 - 1) The valve shall be suitable for use at the location and environment as specified in the specification;
 - 2) The valve shall be designed to prevent the pressure in the pipeline immediately upstream of the valve rising above a preset level;
 - 3) The pressure at which the valve opens shall be adjustable;

- 4) Valve bodies shall be capable of withstanding 1.5 time the rated pressure;
 - 5) Valves shall have a smooth bore to ensure free passage of fibrous or solid material;
 - 6) Flanges to be drilled to BS EN 1092-2:1007 and rated to at least 1.5 times the maximum operating piping system pressure;
 - 7) Valve shall be designed not to cause pressure transients in the pipework; and
 - 8) Materials of construction shall be corrosion resistant and compatible with all fluids being handled.
- (iv) Diaphragm Flow Control Valve (for used water or wastewater)
- 1) This shall be Iris type diaphragm control valve suitable for air flow control application as specified in specification;
 - 2) The flow control valve shall be electrically actuated, unless otherwise specified;
 - 3) The valve shall be suitable for precise flow control and for used water (wastewater) application;
 - 4) The valve shall be Non-clogging and self-cleaning the segment edges;
 - 5) The valve shall be able to control the flow via centrally closing segment acting as concentric diaphragm. It shall be able to continuously maintain the axial central flow and flow rate by varying the segments opening. Visual indicator shall be provided to indicate the opening of the segment;
 - 6) Valve shall be highly energy efficient in flow control with low pressure losses and low turbulences; and
 - 7) Unless otherwise specified, the valve shall be stainless steel grade 316 with bronze hard Chrome plated for the segments.
- (d) Material of construction shall be selected to avoid electrolytic corrosion and erosion.
- (e) When specified for positive operation and remote indication, provision shall be made for fitting and adjusting the set pressure by means of shredder type valve unit and an external pressure reading device to monitor the internal pressure settings.
- (f) Each unit shall have a switch for responding to failure mode so that the signal can be transmitted to an alarm state if specified.
- (g) Limit switches shall be of the enclosed type and protected against the ingress of moisture.

G4.1.27 Ball Float Valves

- (a) Ball float valves shall conform as applicable to BS 1212 Parts 1 to 4.
- (b) Ball float valves shall be designed for installation on the inlet pipe to a storage tank and shall automatically shut off when the water reaches a predetermined level. They shall be of the single beat type with balancing piston, resilient seating and direct float and lever operation.

- (c) The shut-off point of the valve shall be adjustable.
- (d) Ball float valves shall be suitable for long term operation in a partially-open state.

G4.1.28 Plug Valves

- (a) Plug valves shall conform as applicable to BS 1552, BS 5158, BS 5353 and BS 6675. Plug valves designed to BS 5158 and BS 5353 shall be tested to BS EN 12266 Part 1.
- (b) Plug valves shall be of the stainless steel, cast iron or cast steel regular pattern, parallel plug type complying with BS 5158.
- (c) Plug valves shall be flanged to BS EN 1092 or screwed to BS EN 10226-1, BS EN 10226-2, and BS EN 10226-3 to suit pipe diameter.
- (d) Valves below DN125 shall be operated by a wrench or hand wheel directly on the valve spindle. Larger valves shall incorporate the use of gearboxes.
- (e) Plug valves shall be of the non-lubricated type with completely resilient rubber covered and shall be designed for low operating torque.
- (f) The valve shall have minimum basic components and be of quarter turn operation. It should provide straight through flow in the fully open position.
- (g) The plug valve shall be used for isolating and not for throttling duties.
- (h) The plug shall be spring loaded or otherwise to compensate for wear to assure tightness and to prevent loose seating.
- (i) A directional pointer shall be included on the end of the valve stem to confirm plug position.
- (j) Eccentric plug valves shall conform to the following additional requirements besides all applicable specifications of the standards and codes referred to for the non-eccentric type:
 - (i) Conform to AWWA C517;
 - (ii) Minimum pressure rating of 10 Bar unless otherwise specified;
 - (iii) Valve bodies and plugs shall be of CF8M (316 stainless steel), ductile iron or cast iron material depending on the intended service fluid and pressure conditions;
 - (iv) Plug elastomers to be bonded according to ASTM D429, Method B as per AWWA C517;
 - (v) End flanges shall be integral with valve body;
 - (vi) Upper and lower bearings shall be of stainless steel SS316 material of construction, replaceable and permanently lubricated;
 - (vii) Seats shall be minimally 3 mm thick with 95 % nickel content;
 - (viii) Stems shall be gland type multiple V-ring packing, field adjustable and replaceable without valve assembly;
 - (ix) Wrench operated eccentric plug valves shall be capable of being converted to worm gear or automated operation without removing the bonnet or plug from the valve. All wrench operated valves shall be equipped with a 50 mm square nut for use with removable levers or extended 'T' handles;

- (x) Worm gear operators, where required, shall be of the heavy duty construction with the ductile quadrant supported on the top and bottom by oil impregnated bronze bearings. The worm gear and shaft shall be manufactured of hardened steel and run on high efficiency roller bearings. All worm gear operators shall be sized for bi-directional shutoff at the valves design pressure rating; and
- (xi) Each valve shall be given a hydrostatic and seat test with test reports conforming to the requirements of AWWA C517-09.

G4.1.29 Diaphragm Valves

- (a) Diaphragm valve shall conform as applicable to BS EN 13397 and shall be tested in BS EN 12266 Part 1.
- (b) The valve shall be of the non-rising spindle type and shall be double flanged in accordance with BS EN 1092.
- (c) The valve body shall be of a two-piece design, to facilitate the easy removal of the diaphragm.
- (d) For steel or ductile iron pipework, the body shall be in cast iron, pressure rated to at least 1.5 times the maximum operating piping system pressure and where used for water shall be corrosion resistant. For use with abrasive material the body shall be suitably lined.
- (e) The flexible diaphragm material shall be suitable for use with the fluid conveyed and where used for abrasive material, shall be selected for maximum resistance to abrasion.
- (f) Valves shall be suitable for tight shut-off applications.
- (g) Valves specified for use with acid shall be of the 'weiring' type and shall be manufactured from cast spheroidal graphite iron with a perfluoroalkoxy (PFA) lining. External parts of the body shall be coated in a phosphate primer and alkyd paint finish. Diaphragms shall be manufactured from polytetrafluoroethylene (PTFE) faced butyl rubber. Bonnets shall be sealed.
- (h) Valves specified for use in wastewater application or for other abrasive materials shall be of the 'full bore' or 'straight through' type. The body shall be fabricated from cast spheroidal graphite iron, with a soft rubber lining and diaphragm. External surfaces shall be coated in a phosphate primer and alkyd paint finish.
- (i) Valves specified for use with polymer, urea and phosphate shall be of the 'full bore' or 'straight through' type. The body shall be fabricated from cast spheroidal graphite iron, with a polypropylene rubber lining. The diaphragm shall be fabricated from a butyl-rubber blend or similar and approved. External parts of the body shall be coated in a phosphate and alkyd paint finish.

G4.1.30 Ball Valves

- (a) Metallic ball valves shall conform as applicable to BS ISO 7121 and BS EN ISO 17292.
- (b) Thermoplastic ball valves shall conform as applicable to BS EN ISO 16135, BS ISO 9393 Parts 1 and 2, BS EN 917:1997, BS EN ISO 1167-1 and BS EN ISO 1167-2.

- (c) Multi-piece bodies shall be used where it is permissible to work on the ball and seats in situ. Where good practice dictates that the valve needs to be removed, or leakage paths need to be reduced, the one-piece body type shall be utilised.
- (d) The seating material shall be selected for long life, erosion and corrosion resistance.
- (e) The ball support shall be of the floating ball or trunnion type. For applications where line pressure is insufficient to ensure a positive leak tight seal built-in seat loading, specially shaped seating to obtain a designed deformation (ensuring a seat loading force) shall be utilised.
- (f) The valves shall be tested to BS EN 12266 Part 1.

G4.1.31 Hydraulically Actuated Ball Valves

- (a) A single valve manufacturer shall be made responsible for supplying the hydraulically actuated ball valves, the respective packaged power systems and for co-ordination of design, assembly, testing, and installation of each valve; however, this does not relieve the Contractor of the responsibility to comply with the requirements in this Section. Each valve shall be produced and assembled by the manufacturer at a facility owned or operated by the manufacturer and under the direct supervision and control of the manufacturer.
- (b) The Contractor and selected valve supplier shall ensure a complete package consisting of all required hydraulically actuated ball valves, actuators, the hydraulic oil reservoir system, accumulator system, hydraulic control system and the solenoid control system.
- (c) All associated equipment, instruments, electrical devices, components, piping and valves and auxiliary appurtenances required for a completely integrated package shall be included.
- (d) The Contractor shall also be responsible for all interfacing required to integrate the hydraulically actuated ball valves into the influent pumps discharge piping system so that the valves will function to serve all their intended purposes during operation.
- (e) All necessary safety features including override provisions shall also be included.
- (f) The valve seat assembly shall be of the design which enables full adjustment and replacement with the valve in situ.
- (g) The valve seat shall be connected to the ball by a mounting ring which is securely attached and pinned into position after the correct setting has been attained. Seat threaded directly on to the iron ball shall not be acceptable:
 - (i) Valve actuator shall conform to the operating requirements of AWWA C507 and shall be designed to hold the valve in any intermediate position between full open and fully closed without creeping or fluttering. Each valve shall have a physical indicator to show the fully open, fully close and intermediate positions.;
 - (ii) All wetted parts of the hydraulic actuator cylinder shall be corrosion resistant;
 - (iii) Hydraulic actuator cylinder rod seals shall be of the non-adjustable wear-compensating type;

- (iv) A rod wiper for removal of deposits inside the cylinder shall be provided in addition to the external dirt wiper;
- (v) The bearings for the ball and body trunnions shall be bronze of dissimilar hardness as per AWWA C507 standard to prevent galling or binding;
- (vi) All the ball valves shall be subject to hydrostatic, shop leakage and performance tests as specified in AWWA Standard C507;
- (vii) Maximum seat leakage allowable shall be as per AWWA C507-18 according to valve seat type i.e. whether resilient seated or metal seated and according to the specific range of differential pressures applicable across the valve;
- (viii) All filters and strainers filtration rating in microns shall be clearly specified;
- (ix) All internal ductile iron surfaces, except finished or bearing surfaces, shall be shop painted as per AWWA C550;
- (x) All exterior steel or cast or ductile iron surfaces, except finished or bearing surfaces, shall be shop painted with one or more coats of Alkyd primer; and
- (xi) Technical specification requirements to be complied with for each of the hydraulically actuated ball valves are as follows:

Requirements for Hydraulically Actuated Ball Valves

No.	Specification/parameter Description	Requirement	Remark
1.0	Type	Hydraulically actuated metal seated	
2.0	Design Code/Standard	AWWA C507-18 Ball Valves, 6 In. through 60 In. (150 mm through 1,500 mm)	
3.0	Pressure rating	1.5 times pump shutoff head or 1.5 times maximum operating pressure, whichever is higher.	Where specified differently in the Particular Specification for specific applications, the Particular Specification shall take precedence.
4.0	Ends Connection	PN rated raised-face flange to BS EN 1092 or Class rated flat-face flange to ASME B16.1.	Preference for BS EN 1092. Contractor to ensure compatibility of connecting pipe flanges. Flanges flatness shall be machined parallel to each other to within 0.005 inch.
5.0	Port	Full bore with clear unobstructed waterway	
6.0	Open/Close status of Valve upon power failure	Fail to Close	
7.0	Major valve components	Body, ball and operating unit (actuator)	

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No.	Specification/parameter Description	Requirement	Remark
8.0	Materials of Construction		
8.1	Body	Cast ductile iron ASTM A 536	
8.2	Ball	Cast ductile iron ASTM A 536	
8.3	Shaft	ASTM A 564 Type 630, H1150 (17-4 PH) stainless steel or other materials approved by AWWA C507	
8.4	Body Seat	Monel fused to base metal	
9.5	Ball Seat	SS 316	
9.6	Mounting ring connecting seat to ball	SS316	
9.7	Bearings	Bronze	
9.8	O-ring / Packing	Buna N	
9.9	Spring pin	Stainless Steel Type 420	
9.10	Actuator cylinder	Non-metallic materials for all wetted parts of cylinders with enclosed operating mechanisms	
9.11	Actuator cylinder rods	Chromium plated stainless steel	
9.12	Cap screws	B8M Type 316	
10.0	Appurtenances		
10.1	Drain opening	Required drilled and tapped drain opening provided with plug on valve body.	The Contractor and valve supplier shall include any other required appurtenance necessary for a complete integrated hydraulically actuated system.
10.2	Vent opening	Required drilled and tapped vent opening provided with plug on valve body	
11.0	Auxiliary Systems and Control Panels		
11.1	Hydraulic Reservoir System	Consisting of Nitrogen gas vessel, Primary Reservoir Tank, Standby Reservoir Tank, instruments, valves, filters/strainers, gear pumps, pipework and all appurtenances as may be required for completeness	The Contractor and valve supplier shall include any other required auxiliary system and panel necessary for a complete integrated hydraulically actuated system.

No.	Specification/parameter Description	Requirement	Remark
11.2	Accumulator System	Consisting of Primary Accumulator Bank, Standby Accumulator Bank, Accumulators, Valving Manifolds	
11.3	Hydraulic Control Unit	Required in a 2-Door Floor mount SS316 electrical enclosure rated to IP65	
11.4	Solenoid Control Panel	Required in a SS316 enclosure rated to IP65	

G4.1.32 Globe Valves

- (a) Globe valves of 50 mm diameter or less shall be of the threaded, bevelled bronze disc, cast iron body, bolted bonnet and hand wheel operated type.
- (b) Globe valves larger than 50 mm shall be of the flanged, bevelled bronze disc, cast iron body, bronze mounted, outside screw and yoke, hand wheel operated type.
- (c) Globe valves shall conform as applicable to BS 1873, BS EN 13789, BS EN 13709 and BS EN ISO 15761.
- (d) Valves shall be tested to BS EN 12266, Part 1.

G4.1.33 Needle Valves

- (a) Needle valves shall be designed to minimise the effect of cavitation.
- (b) Pressure ratings shall be at least 1.5 times the maximum operating piping system pressure.
- (c) Body ends shall be flanged with raised faces and drilled to BS EN 1092.
- (d) Pneumatically operated needle valves shall be equipped with a pneumatically driven servo system, with sufficient torque to close the valve without hydraulic assistance, independently adjustable throttle valves to control the opening and closing speeds, solenoid operated 'open' and 'shut' pilot valves, 'open' and 'close' limit switches, a position indicating potentiometer, and an exhaust air silencer.
- (e) As a minimum, the material of construction shall be as follows:

PART	MATERIAL
Valve Body and Large Plunger	Meehanite Cast Iron
Pitot Valve, Stem Operating Shaft and Linkage Pins	Stainless Steel
Bushes and Small Valve Plunger	Gunmetal LG2 alloy to BS EN 1982

G4.1.34 Telescopic Valves

- (a) Telescopic valves shall be as specified herein and have the characteristics and dimensions for the intended specific purposes.

- (b) The valve shall utilize a low friction seal to mount to engage the slip tube and mount to the flange of the receiving pipe.
- (c) Valve finish quality shall be of mill finish on stainless steel. Welds shall be sandblasted to remove weld burn and scale.
- (d) Materials shall be as follows:

(i) Slip Tube and Bail	Stainless Steel, Type 304L, ASTM A240
(ii) Stem	Stainless Steel, Type 304, ASTM A276
(iii) Anchor Studs, Fasteners and Nuts	Stainless Steel, Type 316, ASTM A276
(iv) Seal	Urethane or Neoprene
(v) Lift Nuts	Bronze ASTM B584
(vi) Pedestals and Wall Brackets	Stainless Steel, Type 304L, ASTM A276
(vii) Operator Housing	Cast aluminium
- (e) The slip tube shall be constructed of minimum Schedule 10 thickness stainless steel pipe. V-notched weirs, U-notched weirs, scum baffles and /or funnel tops shall be provided as necessary to achieve the intended functions for the specific applications.
- (f) The bail shall be constructed of stainless steel and shall be bolted to the stem and welded to the slip tube.
- (g) The telescoping valve shall be provided with a self-adjusting seal system to restrict leakage between the slip tube and the receiving pipe:
 - (i) The seal shall be a one-piece molded urethane seal with an upper lip type seal and an integral companion flange;
 - (ii) If a neoprene seal and separate companion flange is provided, the neoprene seal shall have a minimum thickness of 1/2-inch and the companion flange shall be stainless steel and shall have a minimum thickness of 3/8-inch; and
 - (iii) The companion flange shall be provided with a bolt pattern suitable for attachment to the flange on the receiving pipe.
- (h) A threaded operating stem shall be utilized to connect the operating mechanism to the bail which in turn is attached to the slip tube. On rising stem valves, the threaded portion shall engage the operating nut in the manual operator or motor actuator. On non-rising stem valves, the threaded portion shall engage the nut attached to the bail:
 - (i) The threaded portion of the stem shall have a minimum outside diameter of 1-1/2 inches. Stem extension pipes are not acceptable;
 - (ii) The stem shall be constructed of solid stainless steel bar for the entire length, the metal having a tensile strength of not less than 75,000 psi (5171 Bar);
 - (iii) Maximum L/R ratio for the unsupported part of the stem shall not exceed 200;
 - (iv) In compression, the stem shall be designed for a critical buckling load caused by a 20 kg effort on the crank or handwheel with a safety factor of 2, using the Euler column formula;

- (v) The stem shall be designed to withstand the tension load caused by the application of a 20 kg effort on the crank or handwheel without exceeding 1/5 of the ultimate tensile strength of the stem material;
- (vi) The threaded portion of the stem shall have machine rolled double lead threads of the full Acme type with a 16 microinch finish or better. Stub threads are not acceptable;
- (vii) Stems of more than one section shall be joined by stainless steel or bronze couplings. The coupling shall be pinned and bolted to the stems; and
- (viii) Stems, on manually operated valves, shall be provided with adjustable stop collars to prevent over travel.
- (i) Stem guide shall be adjustable in 2 directions and provided when necessary to ensure that the maximum L/R ratio for the unsupported part of the stem is 200 or less.
- (j) Stem guide brackets shall be fabricated of stainless steel and shall be outfitted with UHMW or bronze bushings.
- (k) Manual operators for telescopic valves shall comply to the following requirements:
 - (i) The valve manufacturer shall select the proper gear ratio to ensure that the valve can be operated with no more than a 40 lb effort.
 - (ii) An arrow with the word "OPEN" shall be permanently attached or cast onto the operator to indicate the direction or rotation to open the valve.
 - (iii) Handwheel operators shall be fully enclosed and shall have a cast aluminum housing:
 - 1) Handwheel operators shall be provided with a threaded cast bronze lift nut to engage the operating stem;
 - 2) Handwheel operators shall be equipped with roller bearings above and below the operating nut;
 - 3) Positive mechanical seals shall be provided above and below the operating nut to exclude moisture and dirt and prevent leakage of lubricant out of the hoist; and
 - 4) The handwheel shall be removable and shall have a minimum diameter of 400 mm.
 - (iv) Crank-operated gearboxes shall be fully enclosed and shall have a cast aluminum housing:
 - 1) Gearboxes shall have either single or double gear reduction depending upon the lifting capacity required;
 - 2) Gearboxes shall be provided with a threaded cast bronze lift nut to engage the operating stem;
 - 3) Bearings shall be provided above and below the flange on the operating nut to support both opening and closing thrusts;
 - 4) Gears shall be steel with machined cut teeth designed for smooth operation;

- 5) The pinion shaft shall be stainless steel and shall be supported on ball or tapered roller bearings;
- 6) Positive mechanical seals shall be provided on the operating nut and the pinion shafts to exclude moisture and dirt and prevent leakage of lubricant out of the hoist;
- 7) The crank shall be cast aluminum with a revolving nylon grip; and
- 8) The crank shall be removable.
- (v) Pedestals shall be constructed of stainless steel. Aluminum pedestals are not acceptable:
 - 1) The pedestal height shall be such that the handwheel or pinion shaft on the crank-operated gearbox is located approximately 900 mm above the operating floor.
- (vi) Wall brackets shall be used to support floor stands where shown on the Drawings and shall be constructed of stainless steel:
 - 1) Wall brackets shall be reinforced to withstand the maximum operating loads mentioned herein; and
 - 2) The design and detail of the brackets and anchor bolts shall be provided by the valve manufacturer and shall be approved by the ENGINEER. The valve manufacturer shall supply the bracket, anchor bolts and accessories as part of the valve assembly.
- (vii) Operators shall be equipped with fracture-resistant clear butyrate or lexan plastic stem covers:
 - 1) The top of the stem cover shall be closed;
 - 2) The bottom end of the stem cover shall be mounted in a housing or adapter for easy field mounting; and
 - 3) Stem covers shall be complete with indicator markings to indicate valve position.
- (l) Electrical actuators for telescopic valves:
 - (i) Electrical actuators shall be according to the section Electrical Actuators herein.

G4.1.35 Dome Valves

- (a) The principle of operation of dome valves shall be such that the dome component closes against an inflatable seat seal. When uninflated, a controlled gap exists between the seat seal and the dome surface. Full shutoff by valve occurs with the inflation of the seal by compressed air which enters the space between the back of the seal face and ring insert thereby causing the seal face to expand onto and around the dome surface periphery. Before valve opening, the seat seal is relaxed through the compressed air pressure reduction and the controlled gap is re-established following which the dome is actuated to move to the open position.
- (b) The valves shall be capable of pressure tight sealing.

- (c) The inflatable seal shall be clamped in place by a spigot piece and external fasteners holding the top plate assembly to the body.
- (d) The valves shall provide unrestricted full bore flow with the dome completely clear of the flow path. The dome shall move between the fully closed to the fully open position or vice versa by a quarter turn of the shaft in opposite directions.
- (e) Where applicable, the dome valve shall be fitted with open/close proximity switches.
- (f) The valves shall have design provisions to be actuated to become fully closed upon detection of compressed air supply failure.
- (g) The valves shall have lubricated-for-life shaft bearings that require no maintenance.
- (h) Materials of construction for dome valves shall be as follows:
 - (i) Body : Cast iron/SS304/SS316 (with PTFE coating where necessary)
 - (ii) Dome : SS316 (with PTFE coating where necessary)
 - (iii) Inflatable seal : Neoprene/Viton/EPDM/Silicone

The exact material for each component shall suit the specific application in that it shall be compatible with the air, corrosion resistant, durable and not be subject to galvanic corrosion. A comprehensive components/parts list with the corresponding materials of construction shall be submitted for approval by the S.O..

G4.1.36 Jet Control Air Regulation Valves

- (a) The principle of operation of jet control air regulation valves shall be flow control by means of an adjustable co-axial element which varies the annular flow area around the element.
- (b) Jet control air regulation valves shall be of highly aerodynamic design with minimal turbulence, low pressure loss and low noise.
- (c) The valves shall be capable for high precision air regulation control and the degree of precision shall be 0.2 % or better.
- (d) Control performance characteristic of the valves shall be linear across 100 % of the entire stroke adjustment range with a gain factor of close to 1.
- (e) The valve actuator shall be direct flange mounted on the valve and according to BS EN ISO 5210.
- (f) Gas-tight closure shall be achieved, making the valves suitable for swing zone operation and process optimisation. There shall be no necessity for any additional manual or automatic shut-off valve.
- (g) The valve control element shall have very low and constant coefficient of drag over the entire control range thereby requiring low drive torque and compact actuator sizes.
- (h) The central control axis with actuator for sensitive control of air supply with 360 deg. free-flow annular orifice shall allow a compact piping configuration having the air flow meter at a distance of ½ diameter upstream of the valve inlet flange.

Alternatively, the flow meter shall be installed at a distance of 3 diameter upstream of the valve inlet flange where sufficient inlet piping length is available.

- (i) Stroke adjustment in the flow direction to regulate air flow shall enable the flow to remain attached to the wall of the valve, thereby facilitating rapid and effective pressure recovery with extremely low pressure loss. As flow in the valve venturi outlet is homogeneous, the first drop leg to a diffuser grid can be located directly downstream of the valve.
- (j) It shall be feasible to install jet control air regulation valve directly in the pipe without the need for pipe reduction or enlargement.
- (k) The valves shall be designed for ambient and gas temperatures from – 40 deg. C. to + 150 deg. C..
- (l) Materials of construction for jet control air regulation valves shall be as follows:
 - (i) Housing and exposed parts : SS316
 - (ii) Journal bearings : Carbon/Teflon
 - (iii) Seals : Viton/PEEK/Teflon

The exact material for each component shall suit the specific application in that it shall be compatible with the air, corrosion resistant, durable and not be subject to galvanic corrosion. A comprehensive components/parts list with the corresponding materials of construction shall be submitted for approval by the S.O..

- (m) For connection, each end of the jet control air regulation shall be furnished with a loose flange.

G4.1.37 Iris Air Regulation Valves

- (a) The principle of operation of iris air regulation valves shall be flow control by means of a variable iris aperture similar to that of a camera shutter.
- (b) Iris air regulation valves shall be of the free, central flow and self-cleaning design with low turbulence and minimum noise.
- (c) The valves shall be suitable for continuous operation and variable duties.
- (d) Air flow regulation characteristics shall be hysteresis-free.
- (e) The valves shall be provided with flushing connections.
- (f) The valve aperture shall be variable from 0 to 100 % and always maintaining a central flow axis.
- (g) Valves with electric actuators shall capable of achieving up to 1200 operations within an hour. Options shall be available for pneumatic actuator, handwheel or lever actuation.
- (h) Where specified, iris air regulation valves shall be integrated with flow measurement.

- (i) Materials of construction for iris air regulation valves shall be as follows:

- | | | | |
|------|---------|---|---|
| (i) | Housing | : | Cast iron/SS316/Zinc free bronze |
| (ii) | Segment | : | Hard chrome plated bronze (Zinc free)/Stainless steel |

The exact material for each component shall suit the specific application in that it shall be compatible with the air, corrosion resistant, durable and not be subject to galvanic corrosion. A comprehensive components/parts list with the corresponding materials of construction shall be submitted for approval by the S.O..

G4.1.38 Automatic Drain Valves

- (a) Unless otherwise specified, automatic drain valves shall be of the float-operated plug type designed to automatically, and continuously drain oil, water and other condensate from the pipework.
- (b) They shall be sized for continuous operation at the maximum stated working pressure.
- (c) Where shown on the drawings, the drain valves shall incorporate a manual condensate drain facility.

G4.1.39 Pinch Valves

- (a) Pinch valves shall be of the full-bore type employing a natural rubber sleeve suitable for use with wastewater, surrounded by a cast iron or aluminium body.
- (b) Valve closure shall be effected by applying compressed air directly to the annulus surrounding the sleeve or by electrical actuation with manual over-ride. Choice shall be based on long life of the sleeve considering the stresses exerted by the actuation.
- (c) Sleeves shall be readily replaceable. Means to detect leak shall be by means of pressure detection by pressure indicators or any other method approved by the S.O..
- (d) An air filter and pressure regulator shall be provided on the airline to each valve.

G4.1.40 Spindles and Caps

- (a) Operating and extension spindles for valves operated by tee key shall be capped.
- (b) Extension spindles shall be circular in section. For valves installed in chambers, extension spindles shall be provided with split bearings rigidly held on brackets spaced no more than 1500 mm apart. For buried valves, the spindle shall be supported inside a protecting tube held on a purpose-made support fixed to the top of the valve and provided with a spindle guide. A cast iron hinged surface box shall be provided for easy access to the valve spindle.
- (c) Extension spindles for remote operation of valves and penstocks shall be complete with guide brackets for bolting to walls at centres not greater than 2 m.
- (d) The spindle shall be supported such that the L/R (L= Length or span between supports, R = Radius of gyration) for the unsupported part of the spindle shall not exceed 200.

- (e) Extension spindles shall be fabricated in stainless steel type 316L of nominal diameter sufficient to transmit the opening/closing torque. Spindles shall have tapered square ends to enter valve caps or muff couplings. Valve caps and couplings shall be secured by hexagonal headed setscrews.
- (f) Extension spindle couplings shall be provided with split pins to prevent pull-out.
- (g) The length of the extension spindle shall be such that the top of the spindle is not more than 225 mm below cover or wellhead level.
- (h) Guide brackets for motor operated valves and penstocks shall be copper alloy bushed.
- (i) Bearings and spindles shall be suitably protected against corrosion. All rising spindles shall rise vertically within a clear plastic cylindrical protective cover, sized to accommodate the full spindle length in the fully raised position.

G4.1.41 Manual Operating Mechanisms

- (a) Tee key operated valves shall be provided with detachable cast iron spindle caps to take the key. One key shall be supplied for every five valves installed, with a minimum requirement of two keys in any one size.
- (b) Hand wheels shall be shaped to give a safe grip without sharp projections, clearly marked with the direction of closing and be fitted with integral locking devices. A padlock and chain will not be an acceptable device.
- (c) All manually-operated valves shall be capable of being opened and closed by one person only, when the specified maximum unbalanced pressure is applied to the valve in use. Under this condition the total force required at the rim of the hand wheel or at the tee key to open the valve from the closed position shall not exceed 300 Newton (150 N each hand). Where necessary, gearing and bearings shall be provided and the hand wheel sized to fulfil this requirement.
- (d) Hand wheel diameter shall not exceed 450 mm.
- (e) Gearboxes shall be totally enclosed oil bath lubricated. Thrust bearings shall be provided in such a way that the gear case may be opened for inspection or be dismantled without releasing the stem thrust or taking the valve or penstock out of service. Oil and grease lubricated gearing, bearings and glands shall be protected against the ingress of dust and moisture.
- (f) Operating mechanisms shall be of the weatherproof type and those parts subject to submergence shall have a degree of protection IP68 to BS EN 60529 at a depth of submergence of 5 m. Where practicable, operating mechanisms shall be fitted with mechanical position indicators clearly visible from the operating position.
- (g) Headstocks of the non-rising spindle type shall each have an index pointer working over a polished and engraved, graduated, open-to-closed position indicator fixed to the side of the pillar. The indicator shall have a robust clear plastic protection plate provided to exclude dust from the mechanism.

G4.1.42 Solenoid Operated Valves

- (a) Solenoid operated valves, 50 mm and larger, shall be of the diaphragm actuated, spring loaded, single seat composition disc, packless globe valve type controlled by a suitable solenoid pilot valve.

- (b) Solenoid operated valves less than 50 mm shall be of the direct acting single integral seat, full pipe area, globe type. They shall be of all bronze construction with tight seating, renewable composition discs and shall be provided with a manual operator.
- (c) The solenoid pilot valve assembly shall be enclosed in a dust and waterproof cover and shall be provided with a manual operator.
- (d) The valves shall have cast iron bodies with non-magnetic stainless-steel stems, renewable bronze seats, synthetic rubber discs, nylon fabric diaphragms bonded with synthetic rubber and stainless steel springs.
- (e) Solenoid operated valves shall be of the normally closed, energised-to-open type, unless otherwise required for the particular service.
- (f) The operating solenoid shall have direct current coils and an integral rectifier for use on a 230 V A.C. supply, unless otherwise specified. The coils shall be encapsulated in epoxy resin.
- (g) Limit switches with voltage-free change over contacts shall be provided for remote signalling of open and closed positions.
- (h) The solenoid control valve shall be of the two way, normally closed type, and shall be suitable for mounting at any angle.
- (i) The solenoid shall be rated for continuous duty.
- (j) A manual override facility shall be incorporated.
- (k) The valve body shall be brass and the internal parts, stainless steel or brass.
- (l) The enclosure shall have a minimum degree of protection IP 67 to BS EN 60529.

G4.1.43 Penstocks

- (a) General
 - (i) Penstock shall be provided as a single complete package with frame, guide rails, deck stand, spindle, universal couplings, bearings, intermediate bearings and bearing housings/ supports, electric actuator along with motors, headstock, gearing, drive bushing, limit switches, position indicator, integral starter, selector switch, supports, painting, labelling, field control panel, emergency stop button, cabling, etc and all necessary instruments required to control the operation of system/equipment to start/stop/switch over to duty from standby mode or vice versa, etc. and all the other necessary accessories and ancillaries required to form complete and functional system of the plant.
 - (ii) Penstock shall be of cast iron or stainless steel or HDPE based construction and shall comply with BS 7775 or AWWA C561-14 except as specified herein. They shall be suitable for either on-seating or off-seating as required by the application. They shall be drop tight at their operating seating pressure. The leakage rate under off-seating or on-seating conditions shall be stated by the Contractor and is subject to approval by the S.O..
 - (iii) All Penstock shall be watertight when closed against the specified head conditions.

- (iv) The inside sealing area of the penstocks shall be at least 1000 millimetres square and four-sided sealing all round at both upstream and downstream, unless otherwise specified.
- (b) Penstock Design
 - (i) The penstock design and materials shall be selected in such a way that lubrication and maintenance works need to be performed only once a year. Designs which require more frequent maintenance or lubrication will not be acceptable. All lubricants used for these penstocks shall be of food grade or and biodegradable.
 - (ii) The penstocks shall be self-contained type i.e. supplied as a factory assembled unit with the frame, door, seals, stiffeners and penstock stem completely assembled and shipped to the site in a 'ready to install' package. The penstock shall then be uncased at site and installed as a complete assembled unit without stripping down into components. This is to ensure that the performance integrity of the penstock remains as factory supplied condition, thus minimizing the influence of the installation process to achieve optimum performance. Penstocks that are supplied in stripped down component condition or required dismantling and reassembling at site are not acceptable.
 - (iii) All anchor bolts, assembly bolts and nuts shall be of ample section to safely withstand the forces created by operation of the gate. Quantity and size of the anchor bolts shall be as recommended by the manufacturer and design of frame with fewer anchor bolts is preferred. All anchor bolts shall be of stainless steel chemical type unless otherwise specified.
 - (iv) All the penstocks shall be rising spindles type with see-through protective cover. If there are any constraints in providing rising spindles penstocks, the Contractor shall seek the approval of the S.O. for the use of penstocks with non-rising spindles. Rising spindles shall be stainless steel SS316L connected to the door and working through a gunmetal nut rotated by the operating gear. Rising spindle penstocks shall be provided with head and foot brackets, or foot brackets only as required.
 - (v) Non-rising spindles, if specified and approved by S.O., shall be rotating in a gunmetal nut in the door shall be of manganese bronze or phosphor bronze. Where length permits, only the screwed portion need be of manganese bronze or phosphor bronze.
 - (vi) Extension spindles shall be supplied with sockets which shall engage penstock spindles, both of which shall be drilled to receive a bolt which shall be secured by a suitable nut. Guide brackets shall be provided where necessary.
 - (vii) Unless otherwise specified, penstocks shall be fitted with a single stem connected to the centre of the gate. For penstocks having width dimension equal to 2 times the height dimension or greater, dual stems shall be provided.
 - (viii) Operating hand wheels where required shall be of sufficient diameter for the duty.
 - (ix) Hand operated weir penstocks shall be lockable in any position.

- (x) Unless otherwise specified, the extension spindles shall be of stainless steel or better material for the entire length with muff coupling of suitable size and strength. Stem protection tubes of clear transparent polycarbonate or other approved material shall be provided.
- (xi) Penstock spindles lengths shall be sized for an extra 20% design margin unless otherwise specified.
- (xii) Unless otherwise specified, hand wheel, crank, or tee key shall be supplied. The maximum operating effort required at one point on hand wheel rim shall be 25 kg when operated against the maximum differential pressure head. This effort shall be with respect to the handwheel diameter size. Hand wheels are to be marked with an arrow to indicate the direction of opening and closing of the penstock. The maximum size of hand wheel shall be 450 mm diameter. The end of the threaded stem shall protrude 100 mm above the hub of the hand wheel when penstock is in fully closed position.
- (xiii) Where necessary, headstocks, pillars, housing, hand wheels, guide brackets, etc. shall be provided and installed in suitable position. All penstocks, fitted with hand wheels or actuators not mounted directly on the penstock, shall be provided with cast iron or stainless steel headstocks. The height of the hand wheel or actuator shall be between 750 mm and 800 mm above floor level.
- (xiv) Penstock gate shall generally be of one piece construction, circular or rectangular or square as specified with horizontal and vertical reinforcing ribs. To provide rigidity between all wedges, a reinforcing rib shall be provided along all sides. Gate shall be designed to operate smoothly.
- (xv) Where identified in the particular specifications for specific applications, penstocks with multiple-leaf gates shall comply to the following:
 - 1) Facilitate flow control and partial or full closure of flow opening in structural channels where single-leaf penstock door would otherwise be unable to due to limited headroom from structural soffit to top edge of door;
 - 2) Facilitate flow control and partial or full closure of flow opening in structural channels having lowered invert elevation downstream of penstock such that single-leaf downward opening weir type penstock would otherwise be unable to be lowered to yield the maximum flow;
 - 3) Allow overshoot and undershot flow regulation control within a single gate structure;
 - 4) Individually removable gate leaves;
 - 5) Materials of construction of parts and components to be the following as minimum or alternative materials guaranteed to be of equivalent or better corrosion resistance:

Components	Materials
Penstock frame / door (leaf)	Marine grade aluminium/SS316
Door guide	UHMWPE
Seal	EPDM

- 6) Available options of face-mounted or embedded side frames;
 - 7) Available options of raised or flat sill;
 - 8) Available actuation options by handwheel, electric, hydraulic, pneumatic or portable actuators;
 - 9) Feasible for integration into new or existing SCADA systems; and
 - 10) Requirements for single-leaf penstock that are given in this specification document which complement and do not contradict the design requirements of multiple-leaf penstocks shall apply also.
- (xvi) Penstock frame shall be of one piece flange-back type construction. Frames shall have adequate strength to prevent distortion in handling and placing and under all conditions of service.
- (xvii) All surfaces of the frame that receives seat facings and guide rails shall be accurately machined.
- (xviii) Couplings shall be adequate to transmit the stresses from one part of the spindle to the other. They shall be threaded inside and shall be keyed for interchangeability with each section of the spindle.
- (xix) Wall brackets for supporting the intermediate guides shall be with heavy reinforcing ribs and slotted hole for the proper alignment of the penstock. The wall brackets and anchor bolts shall be of such strength as to withstand the maximum thrusts under all operating conditions.
- (xx) Gate Guides shall be designed to withstand the total force resulting from the operating conditions specified and wedging action of the gate.
- (xxi) Gate Guides shall be designed to allow smooth operation of the gate.
- (xxii) Gate Guides shall be integral with the frame and shall be designed to prevent any relative movement between the guides and the frame.
- (xxiii) Unless otherwise specified, the gate sealing system should have been tested through a cycle test in an abrasive environment and should show that the leakage requirements are still obtained after 25,000 cycles with a minimum deterioration.
- (xxiv) Gate sealing material shall be pre-tensioned along the inside of the frame so that the gate can slide easily along the seal. No seal adjustment at site shall be permitted.
- (xxv) Spindles shall be of sufficient diameter at the base of the threads to lift the gate against its self-weight, the resistance at the maximum unbalanced water head and the starting force without buckling or permanent distortion. Couplings shall be adequate to transmit the stresses from one part of the spindle to the other. They shall be threaded inside and shall be keyed for interchangeability with each section of the spindle. Spindle threads shall be rolled. Cut threads are not acceptable.
- (xxvi) Intermediate guides for the spindles shall be provided as recommended by the manufacturer. They shall be adjustable in two directions and spaced at required intervals to adequately support the spindles. Provisions for lubrication are required.

- (xxvii) Unless otherwise specified, wall brackets for supporting the intermediate guides shall be provided and shall be of one piece construction with heavy reinforcing ribs and lashed hole for the proper alignment of the penstock.
 - (xxviii) The wall brackets and anchor bolts shall be graded 316 L stainless steel and size shall be determined by manufacturer to ensure sufficient strength to withstand the maximum thrusts under all operating conditions. All others assembly bolts shall be stainless steel 316 as minimum.
 - (xxix) All parts that are subjected to regular wear and tear such as stem nut shall be easily replaceable at site.
 - (xxx) All welding shall be performed with control robotic sequence to avoid heat distortion. The completed unit shall be acid cleaned and passivated.
 - (xxxi) Motorised penstocks shall be opened and closed using electric actuators. The actuators shall be double action or reversing motor type with totally enclosed micro-limit switches to ensure complete protection of the actuator under all circumstances. All motorised penstocks supplied and installed shall be complete with all the necessary wiring and integral electrical control panel.
 - (xxxii) Actuated penstocks and valves shall be complete with an integral mounting flange to BS EN ISO 5210. Penstock actuator torque shall be sized for extra 20% design margin.
 - (xxxiii) Except for plastic gate penstocks, frames and doors shall be capable of withstanding the impact of floating debris. All penstock frames shall be of the continuous (closed top) design. All surfaces shall be self-draining and shall not readily collect rags, grit or faecal matter. All penstocks shall be installed with a flush invert.
 - (xxxiv) The Contractor shall obtain and submit for the S.O.'s approval all design calculations verifying that the penstocks conform to all applicable requirements of BS 7775 or AWWA C561-14.
- (c) Witness Test and Inspection of Penstocks
- (i) All penstocks of 1000 mm opening or larger shall be subjected to witness testing and inspection by the S.O or his representative at the Manufacturer's works. The S.O. reserves the right to attend witness testing and inspection at the Manufacturer's works for those penstocks which are less than 1000 mm opening but with specified heads of 5 m and above.
 - (ii) All penstocks with 4 sided seals shall be subjected to hydrotest at the maximum design head at the Manufacturer's works.
 - (iii) Every motorised penstock subject to witness testing shall be combined tested with its actuator or operational performance test shall be carried out to verify that the thrust is within design limits and that the actuator is correctly sized.
 - (iv) Test certificates shall be submitted for all penstocks.
- (d) Installation of Penstocks
- (i) Installation of all penstocks, guides, wall brackets, and anchor bolts, etc. shall be as recommended by the penstock manufacturer with excellent workmanship. It shall be responsibility of the Contractor to supervise and check that the installation is performed to manufacturer's recommendation.

- (ii) All machined working surfaces, intermediate bearings, guide brackets, spindles, etc. shall be thoroughly cleaned and a heavy coat of grease or other suitable rust resistant coating given. This coating shall be maintained during all periods of storage and installation up to the time of commissioning and take over.
- (iii) Sealing design between the wall and the frame shall be of EPDM gasket or as specified, that is attached to the back of the penstock and housed in a well design recess to prevent extrusion of the seal when compressed against the wall during tightening. It shall be self-adjusting to compensate for wall unevenness.
- (iv) The sealing rings design shall be able to withstand the closing forces between the sealing gate and the wall. Sealing ring shall be flexible but tough. Alternative materials shall be subject to approval.
- (v) During installation, no grouting or additional adjustment and preloading of the seal shall be required at site to optimize the sealing of the gate. The complete penstocks shall be of a compact design, factory tested for water tightness and delivered as one ready-to-install unit with no screws for adjustment.
- (vi) Penstocks shall be fixed in position using stainless steel 316L anchor bolts. It shall be grouted into position in the structures to provide effective closure of the openings, including provision of formwork, packings, and sand/cement grout or concrete incorporating an additive to impart non-shrink characteristics to the grout or concrete. This closure shall be watertight. The additive shall be approved for use in contact with potable water in water treatment application.
- (vii) Following the completion of each penstock installation, the penstocks shall be operated through at least two complete open/close cycles. Limit switches shall be adjusted following the manufacturer's instructions.
- (e) Site Testing
 - (i) All penstocks shall be tested up to the maximum head as specified unless otherwise instructed by the S.O..
 - (ii) The cost of providing the necessary water and electricity for conducting the tests shall be included in the tender price/rates.
 - (iii) Functional Tests: Prior to acceptance of gate installation, each gate shall be operated to demonstrate its ability to operate without overloading, or excessive vibration.
 - 1) Alignment: Test complete assemblies for correct rotation, proper alignment and connection, and quiet operation;
 - 2) Vibration: Test complete assemblies for excess vibration. Adjust or modify as necessary; and
 - 3) Penstock shall operate smoothly. Penstock spindled shall operate without bending or excessive force on the operator under all conditions.
 - (iv) All necessary adjustments and defect rectification shall be carried out as per S.O.'s approval.

- (v) Performance Test: Prior to acceptance of installation, each complete assembly shall be hydrostatic tested for 3 hours continuously under maximum operation conditions. The leakage shall not exceed 1.25 litres per minute per metre of gate periphery under seating condition and 2.5 litres per minute per metre of gate periphery under unseating condition in accordance with ANSI/AWWA C501, unless other specified and approved by S.O..
- (vi) In the event the test fail, it shall be adjusted, realigned, modified and retested until it passes the leakage test. The final test result shall be submitted to S.O. for approval. All the time and cost shall be borne by Contractor for the re-test.
- (f) Penstock - Cast Iron-Based Construction
 - (i) Cast iron based construction penstock shall have the following minimum requirement:
 - 1) All penstock shall conform to BS EN 1561 not less than Grade 220 and AWWA C501, rising stem type, unless otherwise specified; and
 - 2) Minimum Acceptable Casting Thickness for Cast Iron Components shall be 20 mm.
 - (ii) Wall Thimbles
 - 1) Cast iron, one-piece construction, in accordance with ASTM A126, Class B;
 - 2) Cast centre ring or water stop around periphery;
 - 3) Front Flange: Machined, with tapped holes to suit frame attaching studs;
 - 4) Stamp vertical centre lines of metal with word “top”;
 - 5) Provide permanent gasket of uniform thickness between penstock frame and thimble; and
 - 6) For large rectangular thimbles, furnish holes in invert to allow concrete placement under thimble. Space holes so that unvented invert length does not exceed 600 mm.
 - (iii) Frames
 - 1) Cast iron one-piece construction, in accordance with ASTM A126, Class B;
 - 2) Machine contact surfaces;
 - 3) Machine dovetailed grooves on front face, into which seat facings shall be driven and machined;
 - 4) Machine back flange to bolt directly to machine face of wall thimble cast in concrete;
 - 5) For frames with top and bottom wedges, furnish integrally cast pads machines with keyways to receive wedge seats; and
 - 6) Where side clearance is limited, flanged frames may require mounting holes to be drilled through to front face of frame.

(iv) Discs

- 1) Cast iron, one-piece construction with integrally cast vertical and horizontal ribs in accordance with ASTM A126 Class B;
- 2) Machine dovetailed grooves on seating face, into which seat facings shall be driven and machined;
- 3) Wedge Pads: Integrally cast on disc and machined to receive adjustable wedges; and
- 4) Cast a heavily reinforced nut pocket integrally on vertical centreline above horizontal centreline to receive thrust nut.

(v) Guides

- 1) Cast iron, one-piece, in accordance with ASTM A126 Class B designed to withstand total thrust from water pressure and wedging action;
- 2) Machine contact surfaces;
- 3) Length: Sufficient to retain and support at least half disc in fully OPEN position;
- 4) Attach to frame with ASTM A276 or ASTM A193, Type 316 stainless steel studs; dowel to prevent relative motion between guides and frame or cast guides integrally with frame; and
- 5) Securely attach wedge seats to machined pads on guides.

(vi) Wedges and Seat Facings

- 1) Side wedges for all conditions. Top and bottom wedges for unseating heads as necessary to meet leakage requirements;
- 2) Solid cast ASTM B584, Alloy 865 manganese bronze;
- 3) Machine contact surfaces. Key to cast iron pads to prevent rotation or lateral motion;
- 4) Attach wedges to disc with ASTM A276 or ASTM A193, Type 316 stainless steel studs and nuts; and
- 5) Seat Facings: unless otherwise specified, the seating surfaces shall be gunmetal to BS EN 1982-LG2 or bronze to BS EN 12163, hand scraped and securely fixed to frame or door.

(vii) Stems

- 1) 25 mm minimum diameter, ASTM A276, Type 316 stainless steel;
- 2) Threads: ACME type with RMS surface roughness of 1.6 micrometre or less on flanks for manually operated gates and 0.8 micrometre or less on flanks for electrically operated gates. Extend threaded portion of stem 100 mm above operator when gate is in CLOSED position;
- 3) Stem couplings shall be used when stems have more than one section;
- 4) Stem couplings shall be of same material as stem;

- 5) Stem couplings shall be furnished with internal threads that transmit full thrust of stem;
 - 6) Stem couplings shall be held in place on stem with bolts or keys and keyways;
 - 7) Stem couplings shall be of same size and shall be interchangeable;
 - 8) Stems shall be sized so that ratio of unsupported stem length (L) to radius of gyration (r), does not exceed 200;
 - 9) Stem shall withstand in compression, without damage, thrust equal to at least 2–1/2 times rated output of hoisting mechanism, with a 25 kilograms effort applied to hand wheel or crank at the maximum differential head;
 - 10) Electric motor-driven floor stands to withstand at least 1.25 times output thrust of motor installed condition;
 - 11) Cast iron, bushed stem guides, mounted on cast iron brackets, adjustable in two directions and spaced so that L/r ratio does not exceed 200; and
 - 12) Adjustable stop collar for CLOSED position.
- (viii) Stem Covers
- 1) Transparent plastic, vented pipe stem cover and cap; and
 - 2) OPEN/CLOSED designators and with 25 mm graduations on clear Mylar pressure sensitive, adhesive tape, suitable for outdoor application.
- (ix) Flush-Bottom Closure Seals
- 1) Compressible Resilient Seal
 - Attached to bottom of disc with a bronze or stainless-steel bar and bronze or stainless-steel fasteners;
 - Specially moulded shape designed to fit a lip machined on bottom rib of disc;
 - Shaped to produce a wide sealing area on a machined cast iron stop bar, bolted and keyed to gate frame to form a flush invert; and
 - Differential sealing pressure of resilient seal on stop bar shall be variable by adjustment of side wedges on gate.
 - 2) Alternative Closures
 - Solid, square-cornered, resilient rubber seal in place of bottom dovetail facing and wedging devices;
 - Securely fastened to bottom cross member of frame on a stop plate, with a retainer bar and stainless-steel fasteners;
 - Make top surface of seal flush with invert of gate opening; and
 - Machine full length of bottom edge of disc accurately to contact seal when disc is closed.

(g) Stainless Steel based Construction

(i) General

- 1) Stainless steel penstocks shall be of similar design to cast iron penstocks except that all part in contact with fluid shall be of stainless steel as specified.
- 2) The seal material shall be of a material compatible with any corrosive chemical present in the fluid.
- 3) Gate shall be rising stem type unless specified. The slide/door shall not be allowed to close on its own weight. Gate with slide/door hanging by the stem during open position is not acceptable. The gate slide/door shall be in full contact on both the seating and unseating side of the UHMWPE guides throughout full travel of the penstock.

(ii) Frame

- 1) The gate frame shall be constructed of structural members or formed plate welded to form a rigid one-piece frame. The minimum thickness of all members except for seal retainer shall be 6.4 mm. The frame shall be of the flange back design suitable for mounting on a concrete wall. The guide slot shall be made of UHMWPE (Ultra High Molecular Weight Polyethylene). The frame configuration shall be of the flush-bottom type and shall allow the replacement of the top and side seals without removing the gate frame from the concrete.

(iii) Slide/Door

- 1) The slide shall consist of a flat plate reinforced with formed plates or structural members to limit its deflection to $1/720$ of the gate's span under the design head. The minimum thickness of all members except for seal retainer shall be 6.4 mm.

(iv) Guides and Seals

- 1) The guides shall be made of UHMWPE (Ultra High Molecular Weight Polyethylene) unless otherwise specified, and shall be of such length as to retain and support at least two thirds ($2/3$) of the vertical height of the slide in the fully open position;
- 2) Side and top seals shall be made of UHMWPE (Ultra High Molecular Weight Polyethylene) of the self-adjusting type;
- 3) A continuous compression cord shall ensure contact between the UHMWPE guide and the gate in all positions. The sealing system shall maintain efficient sealing in any position of the slide and allow the water to flow only in the opened part of the gate;
- 4) The bottom seal shall be made of EPDM set into the bottom member of the frame and shall form a flush-bottom;

- 5) Sealing system shall be maintenance-free design via self-adjusting seal. Sealing system using wedges and seals secured on the disc/door are not acceptable; and
 - 6) Use of J-Bulb, P-Seal or Crown Seal are not acceptable.
- (v) Stem and Couplings
- 1) The operating stem shall be of stainless steel designed to transmit in compression at least 2 times the rated output of the operating manual mechanism with a 40 lbs (178 N) effort on the crank or hand wheel;
 - 2) The stem shall have a slenderness ratio (L/R) less than 200. The threaded portion of the stem shall have machined cut threads of the Acme type;
 - 3) Where a hydraulic, pneumatic or electric operator is used, the stem design force shall not be less than 1.25 times the output thrust of the hydraulic or pneumatic cylinder with a pressure equal to the maximum working pressure of the supply, or 1.25 times the output thrust of the electric motor in the stalled condition;
 - 4) For stems in more than one piece and with a diameter of 1¼ inches (45 mm) and larger, the different sections shall be joined together by solid bronze couplings. Stems with a diameter smaller than 1¼ inches (45 mm) shall be pinned to an extension tube;
 - 5) The couplings shall be grooved and keyed and shall be of greater strength than the stem; and
 - 6) Gates having a width greater than two times their height shall be provided with two lifting mechanisms connected by a tandem shaft.
- (vi) Stem Guides
- 1) Stem guides shall be fabricated from type 316L stainless steel. The guide shall be equipped with an UHMWPE bushing. Guides shall be adjustable and spaced in accordance with the manufacturer's recommendation. The L/R ratio shall not be greater than 200.
- (vii) Stem Cover
- 1) Rising stem gates shall be provided with a clear polycarbonate stem cover. The stem cover shall have a cap and condensation vents and a clear Mylar position indicating tape. The tape shall be field applied to the stem cover after the gate has been installed and positioned.
- (viii) Lifting Mechanism
- 1) Manual operators of the types listed in the schedule shall be provided by the gate manufacturer;
 - 2) All bearings and gears shall be totally enclosed in a weather tight housing. The pinion shaft of crank-operated mechanisms shall be constructed of stainless steel and supported by roller or needle bearings;

- 3) Each manual operator shall be designed to operate the gate under the maximum specified seating and unseating heads by using a maximum effort of 40 lbs (178 N) on the crank or hand wheel with the handwheel diameter taken into consideration and shall be able to withstand, without damage, an effort of 80 lbs (356 N); and
- 4) The crank shall be removable and fitted with a corrosion-resistant rotating handle. The maximum crank radius shall be 15 inches (381 mm) and the maximum hand wheel diameter shall be 24 inches (610 mm).

(ix) Yoke

- 1) Self-contained gates shall be provided with a yoke made of structural members or formed plates. The maximum deflection of the yoke shall be 1/360 of the gate's span.

(x) As a minimum, the material of construction shall be as follows:

PART	MATERIAL
Frame, yoke, stem guides, stem extension	Stainless steel ASTM A-240 type 316L or as specified
Slide/Door	Stainless steel ASTM A-240 type 316L or as specified, supporting members with stainless steel slide/gate
Side seals, stem guide liner	Ultra-high molecular weight polyethylene (UHMWPE) ASTM D-4020 or as specified
Compression cord	EPDM ASTM 1056
Bottom seal	EPDM ASTM 1056
Threaded stem	Stainless steel ASTM A-276 type 303 MX or 316
Fasteners	ASTM F593 and F594 GR1 for type 304 and GR2 for type 316
Pedestal, hand wheel and crank	Stainless steel 316L or Tenzaloy aluminium
Gasket (between frame and wall)	EPDM ASTM 1056
Stem cover	Polycarbonate ASTM D-3935
Lift nut, couplings	Manganese bronze ASTM B584 UNS-C86500

(h) High Density Polyethylene (HDPE) based construction

- (i) The penstock shall be provided with a vertical moving door to close square-, rectangular-, circular-, or egg shaped openings. Tightening shall be effected without the use of adjustable wedges by means of a special EPDM spherical/lip shaped rubber profile to withstand pressure on both sides.

- (ii) The penstock shall be watertight on both sides withstanding the design pressure without any visible deformation.
- (iii) The maximum allowable leakage rate is within 1% of what is allowed in the standard of the American Water Works Association AWWA C560.
- (iv) To maintain free passage of flow the penstock shall be supplied with flush invert.
- (v) The door shall be of HDPE (high density polyethylene) with a minimum thickness of 30 mm and if necessary it will be provided with stainless steel AISI 316 strengthening profiles running completely into the guides. The door shall be provided with a replaceable EPDM spherical/lip shaped rubber profile for sealing. The frame- or backplate shall be of HDPE with a thickness of 30 mm. The upper part of the frame plate shall be V-shaped for better guidance of the movement of the door preventing effectively wear on the seal and improving the lifetime and function. The guides and the cross bar shall be of stainless steel AISI 316. The cross bar will be U or C-shaped with a minimum thickness of 4 mm to add extra strength and stiffness to the frame. The trapezium threaded stem (spindle) shall be from stainless steel AISI 316. The threaded spindle part will be bolted to the intermediate operation socket with a striking pin. The stop at the underside of the spindle shall be of stainless steel AISI 316 bearing ring with a nut. The bearing house(s) shall be of a wear resistant polyactal or bronze minimum RD 7 according to DIN 1795 with low contents of zinc. The tightening between wall and frame or backplate shall be of neoprene cell rubber tape to be stuck on the frame plate.
- (vi) The length of the extension spindle required between the top of the valve/penstock and the horizontal axis of the handwheel, both in the case of handwheel operated and electric motor actuated valves shall be determined. The spindle shall be supported such that the L/R (L = Length or span between supports, R = Radius of gyration) for the unsupported part of the spindle shall not exceed 200.
- (vii) Extension spindle couplings shall be provided with split pins to prevent pull-out.

G4.1.44 Tilting Weir

- (a) The motorized overflow weir shall come complete with the electric actuator along with motors, headstock, gearing, drive bushing, limit switches, position indicator, integral starter, selector switch, supports, painting, labelling, field control panel, emergency stop button, cabling, etc. and all necessary instruments required to control the operation of system/equipment to start/stop/switch over to duty from standby mode or vice versa, etc. and all the other necessary accessories and ancillaries required to form complete and functional system of the plant.
- (b) The tilting weirs shall utilize self-adjusting seals. Due to the difficulty to access the tilting weirs when they are in service, tilting weirs that utilize adjustable wedges or wedging devices are not acceptable.
- (c) All structural components of the frame and slide shall be fabricated of 316L stainless steel and shall have adequate strength to prevent distortion during normal handling, during installation and while in service.

- (d) Stainless steel shall be mill finished.
- (e) Welds shall be sandblasted to remove weld burn and scale.
- (f) All iron and steel components shall be properly prepared and shop coated with a primer.
- (g) Minimum material of construction shall be as follows:

Components	Materials
Frame, Invert and Retainers	Stainless Steel, Type 316L
Slide, Stiffeners and Pivot Arms	Stainless Steel, Type 316L
Stem	Stainless Steel, Type 316L
Fasteners, Buts and Bolts	Stainless Steel, Type 316L
Diaphragm Seal	Neoprene ASTM D2000 or EPDM
Lift Nuts	Bronze ASTM B584
Pedestals and Wall Brackets	Stainless Steel, Type 316L

- (h) The frame assembly shall be constructed of formed 316L stainless steel plate. The frame design shall allow for mounting directly to a wall with stainless steel anchor bolts and grout as shown on the tender drawings.
- (i) The slide and reinforcing stiffeners shall be constructed of 316L stainless steel plate. The slide shall be designed to have sufficient stiffness to prevent deflection of more than 1/369 of the span. Reinforcing stiffeners shall be welded to the slide and mounted horizontally. Two vertical stiffeners shall be welded on the outside of the horizontal stiffeners for additional reinforcement when necessary.
- (j) The tilting weir shall be provided with a resilient diaphragm seal system to restrict leakage. The diaphragm seals shall be bolted to the frame and to the slide. It shall fully extend when the slide is in the open position and shall collapse when the slide is closed or partially closed position. The seal system shall be durable and shall be designed to accommodate high velocities and frequent cycling without loosening or suffering damage. All seals must be bolted or otherwise mechanically fastened to the frame and slide. Arrangement with seals that are force fit and/or held in place with adhesives are unacceptable.
- (k) Stem and stem guard shall be provided. A threaded operating stem shall be utilized to connect the operating mechanism to the slide. The threaded portion shall engage the operating nut in the manual operator or motor actuator. The stem shall be constructed of solid stainless-steel bar for the entire length, Stem extension pipes are acceptable. The stem shall be threaded to allow full travel of the slide. Stems of more than one section shall be joined by stainless steel or bronze couplings. The coupling shall be bolted to the stems. The stems on manually operated tilting weirs shall be provided with adjustable stop collars to prevent over closing of the slide.

G4.1.45 Stoplogs and Bulkhead Gates

(a) General

- (i) Stoplogs and bulkhead gates shall be made of stainless steel or marine grade aluminium or as specified. They shall be supplied complete with lifting beams, all associated lifting appurtenances and shall be of the size and mounting type shown on the Drawings and specified in the Particular Specifications.
- (ii) In general, stoplog/bulkhead gate mounting frames shall be recessed into channels. Frame parts protruding into the clear channel flow path will not be accepted.
- (iii) Bulkhead gates shall be non-actuated type, unless otherwise specified.
- (iv) All metal to metal contact between bulkhead gates, bulkhead gate frame and guide frame during insertion and removal shall be eliminated.
- (v) Stoplog panels/bulkhead gates shall be able to self-guide along the guide frame from ground level to invert level of the designated locations. The guide frame shall fully extend from its inserted position to removal position.

(b) Materials of Construction

Stop log / Bulkhead gate	Marine grade aluminium or as specified
Stop log / Bulkhead gate frame	Stainless steel type 316L
Lifting Beam	ASTM A36 Hot Dip Galvanized or ASTM A444
Seals	Neoprene or EPDM
Guides	Ultra-High Molecular Weight (UHMW) Polyethylene or equivalent.
Hooks & Fasteners	Stainless steel, ASTM A276, Type 316
Storage Rack	Marine grade aluminium

(c) Stop Log Panels

- (i) Stop log panels shall be of heights suitable for easy handling and shall add up to the overall specified height. Joints, if any shall be made watertight with neoprene or EPDM seals or by welding. Each panel shall have two hooks to engage the lifting beam. Panel surface shall be mill finish.
- (ii) Adequate drainage shall be provided for each log/gate. Stop log panels/bulkhead gates shall be furnished with UHMW guide surfaces to eliminate metal-to-metal contact with slots during insertion and removal.

(d) Seals

- (i) Seals shall be permanently attached to sides and bottom of each panel. Seals shall be formed all around the waterway channel and between each stop log panel. Seals for bulkhead gate shall be designed for bi-directional sealing unless otherwise specified as single-directional.

- (e) Lifting frame and spreader beam
 - (i) Lifting frame or equivalent appurtenance shall be designed to lift the stop log panels/bulkhead gates from the storage rack and vice versa. Due to space constraint, the Contractor shall provide drawings of the lifting frame during tender. Relevant calculation and tested certification of the frame, spreader beam and lifting hoist shall be provided as required by MOM.
 - (ii) During insertion of stoplog panel or bulkhead gate, the lifting frame or equivalent lifting appurtenance shall be lifted to meet the top of the panel/gate and the lifting frame shall self-engage the panel/gate with no guidance from operators.
 - (iii) During removal of stoplog panel or bulkhead gate, the lifting frame shall be lifted down to meet top of the panel/gate. The lifting frame shall self-engage to the panel/gate which can then be removed out of chamber with no personnel entry into the chamber. Stoplog panels and bulkhead gates shall be designed with lifting lugs suited to be used with the self-engaging lifting frame.
 - (iv) The stoplog panels/bulkhead gates and the self-engaging lifting frame shall be oriented to allow the ease of removal and storage.
 - (v) Lifting frame shall be complete with the chain hoist, chain and spreader beam. The lifting frame and the spreader beam or equivalent appurtenance shall be galvanised steel with rust protection coating as minimum and shall be suitable to lift the stop log panels/bulkhead gates safely. Strong durable sturdy wheel with locking device shall be provided and shall be able to sustain the total weight of the lifting equipment structure and the SWL.
 - (vi) All relevant drawings and calculations of the lifting equipment together with the materials shall be submitted to the S.O. for approval before fabrication and installation is carried out. The Contractor shall submit application to MOM and engage an MOM approved person to test the crane and certify that it is safe for use. The P.E. certification and registration with MOM shall be valid until the end of the Defects Liability Period (DLP).
 - (vii) The lifting hooks on hoists shall be provided with a safety catch to prevent slings or lifting apparatus becoming accidentally detached from the hook. The hook shall be capable of rotating relative to the cross head through 360 °. Any nut used in the hook block assembly shall be castellated and locked in position by a split pin. All hooks, chains, lifting slings shall be to BS EN 1677-5 and according to the latest MOM requirement. The hooks, chains, lifting slings, etc shall be suitable for the local outdoor weather condition application and be corrosion resistant.
 - (viii) The hoisting system for stoplogs shall be manually operated whereas that for bulkhead gates shall be electrically operated.
 - (ix) Hoist and chains shall be galvanised steel as minimum and shall be load tested to verify the SWL. It shall fall vertically from the pulley or drum. A single fall of load chain shall not be used. A load chain collection box shall be provided for each chain hoist unit supplied.

- (x) The spreader beam shall be two points centre suspension complete with rust protection coating, rotating hooks, and shackles, with lifting point to suit the lifting points in stop logs and bulkhead gates and the relevant test and operation certificates. It shall load tested and certified by MOM and clearly marked with the SWL on the beam. The lifting beam / spreader or equivalent appurtenance shall be suitable for use with the stoplogs/bulkhead gates as specified in this Contract. All lifting accessories and lifting points to lift the stop logs/bulkhead gates safely shall be provided under this contract.
- (xi) The entire assembly including the spreader beam and accessories associated with the lifting system shall be completely tested for all functions in the presence of the S.O and a licensed tester.
- (xii) When not in use, the spreader beams shall be stored in in the storage rack in a designated place so that they are not damaged.
- (f) Storage Rack
 - (i) When not in use, the stop logs and bulkhead gates shall be stored in their designated storage rack(s). Location of the storage rack(s) shall be fully exposed to Singapore weather in near marine environment and shall be as specified in the Particular Specifications.
 - (ii) Stoplogs, bulkhead gates and the self-engaging lifting frames shall be neatly arranged in their respective storage racks when not in use.
 - (iii) Minimum material of construction shall be marine grade aluminium.
 - (iv) Suitable corrosion protection system shall be proposed to S.O. for approval.
 - (v) Detailed design including the weld design, bolting and securing mechanism for the rack shall be endorsed by P.E. and submitted to the S.O. for approval.
- (g) Weld quality
 - (i) All weldments of bulkhead gate, bulkhead gate frame, guide frame and lifting frame shall be subjected to minimum 100% visual inspection and 10% dye penetrant testing by qualified third party inspector.
 - (ii) Inspection and test results shall be submitted for S.O.'s approval prior to installation on site.
- (h) Factory test
 - (i) Stoplogs and Bulkhead gates shall be factory tested for functionality prior to installation on site.
 - (ii) Factory acceptance test certificates along with test results shall be submitted for S.O.'s approval prior to installation on site.
- (i) Witness Test and Inspection of stoplogs and spreader beam
 - (i) All spreader beams and associated lifting appurtenances shall comply with BS EN 13135 and be subjected to inspection by the S.O or his representative at the Manufacturer's works.
 - (ii) Load Test certificates for the stoplogs/bulkhead gates, the spreader beam and associated lifting appurtenances shall be submitted to S.O. for approval.

- (j) Site Testing
 - (i) All stoplogs/bulkhead gates shall be tested up to the maximum head and the spreader beam shall be subjected to load test unless otherwise instructed by the S.O..
 - (ii) The cost of providing the necessary water and electricity for conducting the tests shall be included in the tender price/rates.
 - (iii) The Contractor shall provide the required water for leakage testing of the stoplogs and bulkhead gates at site.
 - (iv) Functional Tests: Prior to acceptance of stoplogs/bulkhead gates installation, each stoplog/bulkhead gate shall demonstrate the following as minimum:
 - 1) Functional test of the bulkhead gates shall demonstrate that bulkhead gates are self-guiding during the insertion and removal of gates from the inlet chamber and the reliability of self-engaging lifting frame;
 - 2) Hydrostatic test of bulkhead shall demonstrate structural and operational integrity of the bulkhead gates under maximum operating head;
 - 3) Leakage test of the bulkhead gates shall demonstrate within acceptable leakage rates as follows:

For on-seating head pressure < 5 m:	Leakage < 0.5 l/min per m of seal perimeter
For on- or off-seating head pressure $\geq$ 5 m:	Leakage < $0.1 + 0.067 H_{os}$ in l/min per m of seal perimeter where H_{os} is the on- or off-seating head in m.
 - 4) The size of the stoplog/bulkhead gate is suitable for the opening at the desired location; and
 - 5) The lifting operation to lower stoplog panels or bulkhead gate into position or remove them shall be able to be performed safely and smoothly.
 - (v) If the test is not successful, it shall be adjusted, realigned, modified and retested until it passes the leakage test. The final test result shall be submitted to S.O. for approval.

G4.1.46 Flap Valves/Gates

- (a) General
 - (i) The Contractor shall furnish all labor, materials, equipment and incidentals required for the design, fabrication, factory test, delivery, installation and site test of flap valves/gates as specified so that supplied flap valves/gates will be complete in every aspect and ready for operation under full operating range of flowrates, upstream and downstream heads.
 - (ii) The material of construction for the major components of the flap valves/gates shall be stainless steel or flexible rubberised plastic or equivalent unless otherwise specified.

- (b) Submittals specifically required for flap Valves/Gates
 - (i) The following are submittals specifically required for flap valves/gates in addition to the submittals specified elsewhere in the General Specification regarding documentation:
 - 1) Complete listing of parts and components with their respective material thicknesses and materials of construction;
 - 2) Installation drawings showing all details of construction, details required for installation, dimensions and anchor bolt locations;
 - 3) Maximum bending stress and deflection of the flap under the maximum design head (seating head); and
 - 4) Contact particulars of the supplier's/manufacturer's company headquarters and the location of the principle manufacturing facility. Provide the name of the company that manufactures the equipment if the supplier utilizes an outside source.
- (c) Design Requirements
 - (i) Each flap valve/gate shall be designed to withstand the maximum design head with a minimum safety factor of 4 with regards to ultimate tensile and a safety factor of 2 with regards to yield.
 - (ii) Leakage shall not exceed 1.24 litres/min per m of wetted seal perimeter in seating head condition.
 - (iii) The flap valve/gate shall utilise a resilient seal around the perimeter of the opening.
 - (iv) All structural components of the frame and flap shall be fabricated of stainless steel having a minimum thickness of 6.4 mm and shall have adequate strength to prevent distortion during normal handling, during installation and while in service.
 - (v) Finish: Mill finish on stainless steel. Welds shall be sandblasted to remove weld burn and scale.
- (d) Material requirements
 - (i) The minimum required materials of construction for flap valves/gates are given as follows:
 - 1) Frame: Stainless Steel, Type 304L, ASTM A240
 - 2) Flap, stiffeners and hinge arms: Stainless Steel, Type 304L, ASTM A240 /Flexible rubberised plastic
 - 3) Hinge pins: Stainless Steel, Type 304, ASTM A240
 - 4) Seals: Neoprene or EPDM
 - 5) Anchor studs and fasteners: Stainless Steel, Type 316, ASTM A276
 - 6) Seal: Neoprene or EPDM

- (e) Frame
 - (i) Frame design shall be of the flanged back type suitable for mounting directly to a wall with stainless steel anchor bolts and grout or mounting to a pipe flange with stainless steel mounting studs and a mastic gasket material.
 - (ii) The frame shall be constructed of formed stainless steel plate with a minimum thickness of 6.4 mm.
 - (iii) The angle of the flap when seated against the frame shall be between 3 degrees and 7 degrees from the vertical.
 - (iv) Lifting lugs shall be provided on the top of the frame to facilitate installation.
- (f) Hinge arms
 - (i) Hinge arms shall be constructed of formed stainless steel plate with a minimum thickness of 19 mm and shall connect the frame to the flap.
 - (ii) Dual hinge arms shall be provided on all flap valves/gates in excess of 450 mm nominal diameter or width.
 - (iii) Flap valves/gates in excess of 450 mm nominal diameter or width shall have two pivot joints per arm. There shall be an adjustable lower pivot with limited rotation and a fixed upper hinge lug arrangement.
 - (iv) The hinge pins shall have a minimum diameter of 25 mm and shall be constructed of solid stainless steel rod.
- (g) Flap
 - (i) The flap and reinforcing stiffeners shall be constructed of stainless steel plate, rubberised plastic or a combination of these with a minimum thickness of 6.4 mm.
 - (ii) The flap shall not deflect more than 1/360 of the span under the maximum design head.
 - (iii) Reinforcing stiffeners shall be welded or incorporated to the flap.
 - (iv) A lifting lug shall be provided on the bottom of the flap.
 - (v) The manufacturer's name, opening size and maximum head rating shall be etched or cut from a stainless steel plate and welded on the flap.
- (h) Seals
 - (i) All flap valves/gates shall be provided with a seal system to restrict leakage in accordance with the requirements given in this specification.
 - (ii) The seals shall be held in place by a stainless steel retainer and stainless steel attachment bolts. The attachment bolts shall have a minimum diameter of 8 mm.
 - (iii) The seal system shall be durable and shall be designed to accommodate frequent operation without loosening or suffering damage.
 - (iv) All seals must be bolted or otherwise mechanically fastened to the frame. Arrangement with seals that are force fit and/or held in place with adhesives are unacceptable.
 - (v) The seals shall be mounted so as not to obstruct the water way opening.

- (i) Anchor bolts
 - (i) Anchor bolts shall be provided by the flap valve/gate manufacturer for mounting requirement.
 - (ii) The Contractor shall ensure that all civil and structural requirements for flap valves/gates installation are in order prior to installation.
 - (iii) The valve/gate assemblies shall be installed in a true vertical plane, square and plumb.
 - (iv) The Contractor shall fill the void in between the valve/gate frame and the wall with non-shrink grout as shown on the installation drawing and in accordance with the manufacturer's recommendations.
- (j) Field Testing
 - (i) After installation, all valves/gates shall be field tested in the presence of the S.O. to ensure that all items of equipment are in full compliance with this Specification document.
 - (ii) All test data and results shall be recorded, signed and submitted to the S.O..

G4.1.47 Diverter Valves

- (a) General
 - (i) Diverter valves shall be of rugged, heavy-duty, corrosion resistant and totally enclosed construction for abrasive applications.
 - (ii) Dry as well as wet material such as waste water screenings and sludge, etc. shall be able to be handled by the diverter valves without problems.
 - (iii) Actuation of the slide gates or flaps may be by electrical, pneumatic or hydraulic actuators.
 - (iv) Inspection panels shall be removable to facilitate access, maintenance and cleaning of seals and slide gates.
 - (v) Unless otherwise specified, diverter valves shall be of the inverted Y-shaped single inlet and double outlet chute housing configuration.
 - (vi) The diverter valve shall be fully compatible for installation and Integration with the outlet of a conveyor or silo or tank to receive the discharging material as described and illustrated by the particular specifications and drawings.
 - (vii) Diverter valve blade and shaft seals shall be replaceable without dismantling the entire valve from the silo, tank or conveyor to which is connected.
 - (viii) Diverter valve interior construction shall have the option to be fitted with replaceable wear liners for added abrasion resistance where necessary.
 - (ix) Material discharge rate out the diverter shall be equal to the rate of material feed into the diverter to prevent build-up inside the diverter chutes.
 - (x) A variety of connection transition adaptors such as round-to-square, square-to-round, square-to-square including those having differing dimensions of the inlet and outlet cross-sections shall be available with the diverter valve.
 - (xi) Additional features where given in the particular specifications or drawings shall be complied with.

(b) Type

- (i) The following types of diverter valve shall be acceptable depending on the specific applications:
 - 1) Diverter valve with two (2) independently actuated slide gates that allow or block material passage into the respective connecting chutes through the opening or closing of the slide gates.
 - 2) Diverter valve with a single flap rotating about a pivot shaft and blocking material passage for the chute closed by the flap and allowing material passage through the other chute not closed by the flap.
- (ii) Material passage flow through the diverter valves shall be by gravity and the angle of chute inclination shall be at 45 to 70 degrees to ensure minimal adherence of material to the surfaces of the chutes interior.
- (iii) Material flow passage shall be possible through either chute with the respective slide gate opened; or through both chutes with both slide gates open; or regulated through just one chute or both chutes with the respective gate(s) opened partially.
- (iv) Diverter valve using a single flap shall be able to allow material passage through the opened chute or have the flap positioned centrally so that material flows into both chutes.
- (v) The slide gate or flap shall be able to fully close or open on a moving column of material at all times.

(c) Major Parts and Components

- (i) The major components of a diverter valve shall consist of the following as a minimum:
 - 1) Bifurcated chute shaft assembly;
 - 2) Bifurcated chute body;
 - 3) Inlet flange (1 no. per diverter valve);
 - 4) Outlet flange (2 nos. per diverter valve);
 - 5) Actuator (1 no. for flap type diverter valve; 2 nos. for slide gates type diverter valve);
 - 6) Flexible connectors;
 - 7) Electrical, pneumatic or hydraulic connections as applicable;
 - 8) Solenoid valves as required;
 - 9) Proximity sensors and limit switches as required; and
 - 10) Positioners as required for applications requiring apportioning of material flow.

(d) Materials of Construction

The following materials of construction for major components shall be applicable:

- (i) Chute body: Aluminium alloy/SS316L
- (ii) Flap: Polymer composite/mild steel with approved liner/SS316L

- (iii) Slide gate: SS316L
- (iv) Wear liner: Abrasion Resistant (AR) steel/chromium carbide/ceramic

G4.1.48 Electric Actuators

- (a) Each actuator shall be supplied complete with installation instructions and wiring diagram and sufficient spare covers, screws and seals to make good site losses.
- (b) The electric actuator shall be totally enclosed weatherproof electric drive unit, and a totally enclosed gear box that operates the gate / valve /shaft.
- (c) The electric actuator shall be suitable for 400 V, 3Ph, 50-Hz, unless otherwise specified.
- (d) The electrical actuator shall be provided with the following as a minimum:
 - (i) An alternative system for manual hand wheel operation which shall be lockable. Padlocks to be provided;
 - (ii) An interlock, to prevent engagement of the hand wheel whilst the actuator is being power driven and to positively disengage the manual drive when the power drive is started;
 - (iii) Reversing type motor starter complete with isolating switch;
 - (iv) Local and remote-control selector switch when specified which shall be lockable. Padlocks shall be provided;
 - (v) Open, stop and close push-buttons;
 - (vi) Potentiometer, transducer and other necessary devices or electronics for remote valve position indication and control when remote monitoring or control is specified;
 - (vii) Mechanical dial indicator or back lit LCD screen with 0-100% marking to show valve or penstock position;
 - (viii) A monitoring relay with a normally open contact to indicate that the actuator is available for remote operation shall be provided with the relay being energised from the control transformer only when:
 - 1) Local/Off/Remote selector switch is in Remote position;
 - 2) 3 phase power supply available;
 - 3) Control supply available;
 - 4) Phase rotation correct;
 - 5) Thermostat not tripped; and
 - 6) Local stop button not locked off.
 - (ix) Auxiliary and interposing relays as necessary; and
 - (x) Installation and wiring diagram and sufficient spare cover screws and seals to make good site losses.
- (e) The hand wheel shall not rotate during electrical operation. An arrow and either the word OPEN or CLOSE shall be cast or permanently affixed on the hand wheel to indicate the appropriate direction to turn the hand wheel.

- (f) Torque switch shall be provided for mechanical disengagement of the drive at the extremes of valve operation in the event of excess torques.
- (g) Electric actuators shall operate valves/ penstocks at opening rates that will not impose unacceptable surge pressures on the pipework.
- (h) Electric actuators shall be rated at not less than 20 percent in excess of the torque required to operate the valve under maximum working conditions.
- (i) The motor actuator shall include the motor, single or multi stage reduction gearing, reversing starter, torque and limit sensing in a watertight housing with a minimum of IP67 rating.
- (j) The actuator shall comply with the following as a minimum:
 - (i) The actuator shall all be a single or multi-reduction unit consisting of worm-gearing.
 - (ii) The gears shall be of hardened alloy steel and the worm-gear shall be alloy bronze. The design should be such as to permit the gear case to be opened for inspection or disassembled without releasing the stem thrust or taking the butterfly gate out of service.
 - (iii) All power gearing shall be grease or oil lubricated in a sealed housing. The gear case will permit the actuator to be mounted and operated in any position. Ball or roller bearings shall be used throughout. The combined gear ratio shall ensure “self-locking” characteristics at all times and shall be sufficient high ratio as to inhibit “back driving” the actuator.
 - (iv) Actuator enclosures shall have a minimum protection IP67 to BS EN 60529. The enclosures of actuators installed in chambers or other locations where flooding is possible, such as basements and pipe galleries, shall have a minimum protection IP68 with sealing O-rings.
 - (v) Unless otherwise specified, actuator which are inaccessible or 2 m above finished-floor level or install inside any bunded facility shall have a LCS/LCP for ease of local control.
- (k) The motor of the actuator shall comply the following as a minimum:
 - (i) The motor shall be of the totally enclosed, non-ventilated, high-starting torque, low-starting current type for full voltage starting. The supply power for operation shall have Class F insulation and a motor frame designed specifically for use in motor operated actuators.
 - (ii) The actuator motor must attain full speed before stem load is encountered.
 - (iii) Motor construction shall incorporate the use of stator and rotor as independent components from the butterfly gate operation such that the failure of either item shall not require actuator disassembly or gearing replacement.
 - (iv) The actuator shall be suitably sized, amperage rated, reversing starter. A control power supply unit shall be included to provide a 24V DC source.
 - (v) The actuator shall include a phase discriminator or automatic correction to prevent against improper phasing. Diagnostic information shall be available both from an actuator-mounted display window and remote.

- (vi) Actuator electric motors shall comply with BS EN 60034 or IEC 34. For non-modulating type actuators, the motor short-time rating (STR) shall permit the successive full travel operation of the travel from open to closed and vice versa but shall be not less than 15 minutes for standard S2 duty configuration. For quarter-turn valves, it shall be 3 to 4 s per inch whereas for multi-turn valves, it shall be 5 s per inch. For modulating type actuators, the motor shall have a duty-type rating (DTR) S4 as minimum to meet varying cyclic load requirements of the valve.
- (vii) All bearings of the motor shall be of the ball or roller type and thrust bearings shall be provided where necessary. All bearings shall be provided with suitable seals to confine the lubricant and prevent the entrance of dirt and dust.
- (viii) Electric motors shall be provided with built-in thermal protection complying with BS EN 60034-11. Thermostatic protection of the actuator shall be via a thermostat embedded in the motor winding and shall be of the automatic reset type.
- (ix) Each actuator shall have a built-in starter suitable for 60 starts per hour, and shall comprise mechanically and electrically interlocked reversing contactors of appropriate rating to the motor size with coils fed from a 120V control transformer of 40 VA minimum rating. The common connection of the contactor coils at the transformer shall be earthed so that the contactors drop out in the event of leakage to earth. The primary winding shall be separated from the secondary by an earthed screen and the secondary output shall be protected by an easily replaceable cartridge type fuse. There shall be no solid-state components incorporated in the actuator to be installed outdoor or if its power or control cable is installed outdoor.
- (l) The wiring for the actuator shall comply the following as minimum:
 - (i) All internal wiring shall be tropical grade PVC insulated stranded cable of 5A minimum rating for control circuits and of appropriate size for the 3-phase power circuits.
 - (ii) Each wire shall be clearly numbered and identified at each end. The terminals shall be of the stud type. Cable entries shall be suitable for PVC/SWA/PVC cables. Cable gland shall be consistent with the IP ratings given for the actuator. Motor wiring shall be internally connected to the terminal compartment for customer access. All conduit connections shall be watertight.
 - (iii) The terminal compartment shall be separated from the inner electrical components of the actuator by means of a watertight seal and the compartment shall be provided with a minimum of three threaded cable entries. All wiring supplied as part of the actuator shall be contained within the main enclosure for physical and environmental protection. External conduit connections between components are not acceptable. A durable terminal identification card showing plan of terminals shall be provided attached to the inside of the terminal box cover indicating serial number, external voltage values, wiring diagram number and terminal layout.
 - (iv) All wiring for external connections shall be connected to marked terminals. Three conduits as minimum shall be provided in the enclosure to bring power and control wiring into a separately sealed terminal compartment.

- (m) Each actuator shall be provided with both open and close torque protection and position limit switches and shall comply the following as minimum:
 - (i) At least one change over contact at each end of travel shall be available for remote indication and interlocking.
 - (ii) The torque switch shall be adjustable (40 percent to 100 percent of rated torque) and shall be adjustable in 1 percent increments. "Latching" to be provided for the torque sensing system to inhibit off during unseating or during starting in mid-travel against high inertia loads.
 - (iii) Torque switches for mechanical disengagement of the drive at the extremes of valve operation in the event of excess torque.
 - (iv) The limit sensing mechanism shall be contactless with a setting range of 2.5 to 100,000 turns with a resolution not to exceed 15 degrees of one actuator centre column revolution. Absolute encoder shall be used where appropriate in terms of more technologically advanced features than the limit sensing mechanism.
 - (v) Calibration of the torque or limit sensing should be achieved without the removal of any actuator enclosure covers. The actuator should be capable of being fitted with the correct number of contact outputs required to produce the operation specified unless an absolute encoder is used.
 - (vi) In the event of Volt-free changeover type contacts being specified, these shall comply the following as minimum:

The following data signals shall be provided:

 - 1) Valve /Penstock fully open;
 - 2) Valve /Penstock fully closed;
 - 3) Valve /Penstock operating;
 - 4) Power Supply failure; and
 - 5) Remote control available.
 - (vii) Monitoring relay will provide feedback for motor trip and power supply failure. Monitoring relay also providing feedback for local/off control. The rating of voltage-free contacts shall be not less than 2 A at 120V AC or 24V DC unless otherwise specified. The contacts shall be suitable for inductive load switching.
 - (viii) Open and close torque and position limit switches with separate changeover volt-free contacts at each end of travel for remote indication and inter-locking. Means shall be provided to prevent the open torque switch tripping while the initial unseating hammer blow effect is applied.
 - (ix) Unless specified specification, two extra sets of limit switches with at least three switches per set, each set independently adjustable to any valve/penstock opening or closing position shall be provided for redundancy.
 - (x) Anti-condensation heater shall be provided. Otherwise, actuators in IP68 enclosures shall be provided.
 - (xi) A separate or segregated terminal box shall be provided for the connection motor, heater and control cables.

- (xii) Actuators installed outdoors shall not contain electronic components unless these are necessary to achieve specified functions.
- (n) The control of the electric actuators shall be accordance with the instrumentation, control and automation specification.
- (o) Conventional 24 Vdc interface signals shall generally be used.
- (p) Command Set: As a minimum, the valve will be capable of accepting the following commands:
 - (i) Valve /Penstock Open (maintained contact); and
 - (ii) Valve /Penstock Close (maintained contact).
- (q) Feedback: As a minimum, provide the following status information as potential-free contacts:
 - (i) Opened limit reached;
 - (ii) Closed limit reached;
 - (iii) Valve / penstock fault; and
 - (iv) Ready for remote operation.
- (r) Each actuator must be performance tested and individual test certificates shall be supplied with the actuator free-of-charge. The test equipment shall simulate a typical valve/penstock load and the following parameters should be recorded:
 - (i) Current at 100 percent, 25 percent torque setting and at stall;
 - (ii) Torque at 100 percent, 25 percent torque setting and at stall;
 - (iii) Test voltage (V AC, phase Hz);
 - (iv) Actuator output speed or operating time; and
 - (v) In addition, the test certificates shall record details of specification, such as gear ratios for both the manual and automotive drive, closing direction, and wiring diagram code number.
- (s) Power Disconnect:
 - (i) Location: Integral with or adjacent to the electric actuator but isolated from actuator terminal compartment; and
 - (ii) Mechanism: Meet applicable requirements of Article Nonfused Switch Disconnectors, Individual, 0 to 600 Volts in the General Specification section on Basic Electrical Materials and Methods.
- (t) Enclosure:
 - (i) Actuator enclosures shall have a minimum protection IP 67 to BS EN 60529 and suitable for the installation in hazardous location as per Hazardous Area Classification as specified in the specification; and
 - (ii) The enclosures of actuators installed in chambers or other locations where flooding is possible, such as basements and pipe galleries, shall have a minimum protection IP68 with sealing O-rings unless otherwise specified.

G4.1.49 Pneumatic Actuators

- (a) Pneumatic actuators shall be provided as specified in the Particular Specifications. Actuator shall be designed allowing installation in any possible position.
- (b) The valve actuator mechanism coupled to the cylinder shall be totally enclosed. The cylinder shall be rigidly secured to the mechanism housing and shall not pivot, rotate, or swing during operation. The piston rod shall be enclosed in the mechanism housing and shall not be exposed to view. Cylinders shall provide satisfactory operation using dry, oil-free air at 5.5 bar g pressure or as specified and shall be factory tested. The certified test report and certificate shall be submitted to S.O. for approval.
- (c) Pneumatic actuators shall be fitted with a local valve position indicator with output signal proportional to valve position and hand wheel operated manual override if specified. Where pneumatic actuators are used on fail close valves, means shall be provided for automatic closing on supply air failure.
- (d) Where specified, pneumatic actuators shall have a separate local control station (LCS).
- (e) The air connections on each actuator shall be equipped with an adjustable flow control valve, air filter and pressure regulator as minimum. Material shall be 316 stainless steel unless otherwise specified
- (f) The flow control valves and connecting piping shall be arranged to permit control of the flow rate of exhaust air from the cylinder and to permit independent adjustment and control of valve opening and closing speeds. Control valves shall be sized so that the time required for the piston stroke cycle time is adjustable between 30 and 120 seconds. Flow control valves shall be manufactured from brass or stainless steel.
- (g) The actuator shall be designed to eliminate any risk (even in case of damage) of contact between the actuator lubricant and the fluid conveyed.
- (h) Pneumatic actuators shall have conventional 24 Vdc i/o signals, in accordance with the P&IDs.
- (i) Modulating type pneumatically actuated valves shall have the following additional accessories:
 - (i) Actuator shall be designed allowing installation in any possible position;
 - (ii) An electronic or digital positioner which can accept 4-20 mA input signal and position the valve accordingly;
 - (iii) Intelligent positioners shall be provided with an electronic position transmitter to provide 4-20 mA HART feedback regarding the position of the valve and status of the positioner;
 - (iv) Pressure gauges of appropriate ranges in the supply and signal air lines; and
 - (v) Limit switches to indicate fully open and fully closed positions.

G4.1.50 Hydraulic Actuators

- (a) Hydraulic actuators shall consist of a hydraulic system complete with oil reservoir and oil gauge. The hydraulic pumping system shall be robust and of adequate capacity to smoothly operate the valve or penstock. Apart from the electric motor drive for the pump, a hand operated drive or pump shall be provided by means of which the hydraulically actuated valve or penstock could be operated should the need arise. The following shall be provided as a minimum:
 - (i) Limit switches shall be provided to give indication of valve or penstock closed, open and moving and also to start and stop the hydraulic oil pump.
 - (ii) The oil pump stop limit switch shall be triggered when the valve or penstock is in the fully open position. The event of the valve or penstock closing due to mechanical failure shall be indicated by limit switch. An annunciator shall indicate when the valve or penstock drops below its mid position. The remaining limit switches shall be effected either by starting the oil pump or by operating the hand pump.
 - (iii) Where specified, the valve or penstock shall be provided with counter weights to close (or open, as the case may be to make it failsafe) the valve or penstock in case of an electrical power failure.
 - (iv) A self-closing regulation shall be provided which shall allow the valve or penstock to reach 80 % of the closed position quickly and the rest at reduced speed to prevent slamming of the valve or penstock.
 - (v) The solenoid valves used in the system shall be of approved high-quality design suitable for continuous operation arranged to close when energised.
 - (vi) Remote indications shall be provided on the hydraulic power pack for valve or penstock for the statuses of closed, open and moving if it is an open-close application or the actual percentage opening or close in case of a regulating application. For regulating valves or penstocks this shall be achieved by means of a position transmitter, which shall provide a 4-20 mA HART output corresponding to the position. Provision shall be made to operate the actuator manually from this panel.
 - (vii) The limit switches provided shall be flood proof.
 - (viii) Each hydraulic power pack (Hydraulic Power Unit, or HPU) shall incorporate its own PLC which shall comply with the requirements for complex machine PLC's as specified in the relevant sections of the General Specification G6 and Particular Specification P6 for the Instrumentation, Control and Automation discipline.
 - (ix) Each hydraulic power pack (Hydraulic Power Unit, or HPU) control panel shall provide, as a minimum, the interface signals shown on the P & IDs and as specified in the Particular Specification P4 Mechanical Works for the concerned contract works.
 - (x) Each hydraulic power pack (Hydraulic Power Unit, or HPU) control panel shall have two power supply input connections:
 - 1) For normal three-phase supply to power hydraulic pump(s); and

- 2) For single-phase UPS supply to power HPU PLC and control components only.

G4.1.51 Valves for Cold Water Plumbing

- (a) General
 - (i) All valves shall be of approved manufacture and shall be installed in positions where they are easily accessible for maintenance and operating purposes.
 - (ii) All water valves shall be selected for operating pressure of not less than two times the expected pressure in the system and shall have been tested to at least twice the operating pressure.
 - (iii) For working pressure up to and including 1000 kPa, all valves of 50 mm nominal bore and larger shall be flanged. Valves operating at this pressure but below 50 mm nominal bore shall be screwed. For working pressure above 1000 kPa, all valves shall be flanged.
- (b) Construction
 - (i) All valves shall have bronze valve faces and seats and manganese bronze spindle.
 - (ii) Valves installed in ferrous piping up to and including 75 mm nominal bore and in all copper piping (if any) shall be of bronze bodies.
 - (iii) Valves installed in ferrous piping above 75 mm nominal bore shall have cast iron bodies.
- (c) Sluice Valve
 - (i) Sluice valve shall be provided on pipes 75 mm diameter and over and shall conform to BS 5163-1, BS 5163-2, BS EN 1074-1 and BS EN 1074-2 rated at 16 bar.
- (d) Draw-off Tap and Stop Valve
 - (i) Draw-off tap and stop valve shall be provided on pipes smaller than 75 mm diameter.
- (e) Float Ball Valve
 - (i) Float valve shall be either piston type conforming to BS 1212: Part 1 or diaphragm type conforming to BS 1212: Part 2.
 - (ii) Floats shall be copper or bronze type and shall conform to BS 2456.
 - (iii) Floats for ball valves shall confirm to BS 1968.
- (f) Check Valve
 - (i) Check valve shall be of non-slam type and shall conform to BS EN 12288.

- (g) Pressure Reducing Valve
 - (i) The valve shall be a hydraulically operated, diaphragm-actuated globe or angle pattern valve. It shall be pilot-controlled by a direct-acting, adjustable, spring loaded, normally open diaphragm valve which closes when downstream pressure exceeds the spring setting. The valve shall have a cast iron body with bronze valve faces and seats. It shall have a maximum pressure rating of 20 bar.
- (h) Strainers
 - (i) Strainers shall be Y-type with removable stainless-steel baskets. All strainers shall be provided with isolating valves arranged so that the strainers can be cleaned without draining the system.
 - (ii) Where a system cannot be shut-down for strainer cleaning, a valved bypass shall be installed in parallel with the strainer.
- (i) Pressure Gauges
 - (i) Pressure gauges shall be installed on the inlets and outlets of pressure reducing valves and where indicated on the Drawings.
 - (ii) Pressure tapings for the gauges shall be complete with cocks and screwed connections.
 - (iii) Pressure gauges shall be of the Bourdon tube type with 100 mm diameter direct reading dial face and stainless steel casing. The dials shall be calibrated in SI units (kPa) and shall be accurate within 0.5 kPa. Each division of the scale shall not be more than 1 kPa. Pressure gauges shall be suitable for the range of pressure expected and the indicator shall be in the mid-range of the scale during normal use.

END OF SECTION